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Faculty of Engineering & Physical Sciences

MSc Data Science Dissertation

Mitigating Echo Chamber Issue in Movie Recommendation System

A project supervised by Dr. Lee Gillam

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ABSTRACT

In the digital age, there has been a huge need for the movie recommendation systems. The Echo Chamber issue has become a major problem for movie recommendation algorithms since it frequently limits the users to a loop where they get recommended with the similar type of contents. The main objective of this project is to mitigate the echo chamber issue in the movie recommendation system. In movie recommendation system the echo chamber issue occurs when users are consistently recommended with the similar type of movies. To mitigate the echo chamber issue, in this project different variations of the recommendation models is proposed to find the balance between recommending the similar type of movie according to the user's preference along with ensuring the diversity in the recommendations. The research proposes a novel approach that combines the strengths of content-based filtering and collaborative filtering methods to address the echo chamber issue. By using the advantages of these models, the proposed methods aims to provide users with the more personalized recommendations along with an element of diversity and surprise in the recommendations. In conclusion the proposed project offers a thorough background solution to mitigate the echo chamber issue in the movie recommendation systems. By seamlessly blending personalization with diversity, the proposed approach ensures the system implies more improved user experience, ensuring that the users are continually introduced to a wide range of movies

Keywords : Echo chamber, Content-based filtering, Collaborative filtering, Hybrid, Diversity, Genres, Filter bubble, recommendations, mitigating.

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1 Introduction

The immense amount of data available in the huge digital world of the 21st century is both an advantage and a challenge. As we have a large amount of data in our hand, it will be hard to filter out the important and essential data from the ocean of data available. The availability of huge amounts of data, known as information overload, has resulted in the development of systems that can filter the data and produce choices that are specific to the user preferences. This is where the Recommender Systems comes into play[1][2].

1.1 Background

1.1.1 Recommendation System

The Recommendation system is an advanced software tools and applications, designed to provide customised recommendations to the user. From choosing which book to read, movie to watch, or a product to buy, the recommendation systems are important in introducing users toward choices that will suit their requirements and preferences[1][2]. The recommendation systems have become an important part of many online platforms from e-commerce to video streaming services. The system focuses on particular categories like movies, music, or products, to provide effective suggestions for that specific category[3]. The system has a very strong foundation which includes understanding the user's preferences, either by analysing their previous behaviour or by comparing them with the similar users. Over time, the logics and algorithms in the recommendation system has evolved, improving its accuracy and effectiveness. At first, the recommendation systems were simple, based on the basic statistical data and basic user preferences. But over the years, the development of machine learning algorithms and availability of huge data, these recommendation systems have developed to handle more complex user behaviour[1]. The recommendation system has a huge influence in the real world. For instance, major e-commerce organisations like Amazon largely depended on the recommendation system to recommend products to the users according to their preferences, to increase their sales and profitability. To make sure that user enjoys the content and spend more time on the platform, streaming platforms like Netflix, Spotify and, YouTube uses recommendation systems to recommend contents according to the user's taste. Even e-book platforms like Kindle uses the system to recommend books, making sure the recommended contents are relevant to their preference and interesting[2]. Like most of the technologies, recommendation systems too have few challenges. One of the important issues is the "Echo Chamber" or "Filter Bubble" issue, where users get recommended with the contents that is very similar with the contents that they have already consumed. This leads to very limited perspective and exposure to different contents.

1.1.2 Movie Recommendation System

In today's world of digital streaming, where a huge amount of data is available in our hands, the main challenge is not just to recommend contents, but also to recommend contents that match with the user's preference. The Movie recommendation was developed to solve this challenge, which recommends movies related to their tastes. The movie recommendation system works based on two main parts: users and items. The main objective of the system is to predict the user's rating or preference for a movie. This system is used in the movie streaming platforms like Netflix and Amazon Prime to make sure that the users are very well engaged with the contents and spend more time on the platform, which will increase the platform's revenue. The movie recommendation system acts as a bridge between the user and the contents. The system analyses the user's previous behaviour and recommends the contents that the user will like[4]. To analyse the user's preference the system collects the data of the user's viewing history, search queries, ratings, and reviews[5]. At first the system collects these data about the user and then the algorithms start analysing these data to find the patterns, trends, and correlations of users. Based on the identified patterns and trends the system predicts the movies that the user will enjoy[5][6]. After recommending the movies, the system observes the user's reaction to the recommendations. Is the recommended movie being watched? Is the recommended movie rated or not? By collecting furthermore reaction data the system will analyse it and fine tunes the recommendations[6]. The initial challenge of the system is the cold start problem, where the system has to provide recommendations to the new users without any historical data of the user. As the number of users and movies keeps on increasing, the system must ensure that the recommendations are accurate[5]. Also, as the system recommends movies as per the user's preference, it is important to ensure that the recommended movies are diverse, allowing the users to explore new genres and not getting stuck in the echo chamber[6].

1.1.3 Echo Chamber Issue

The term "Echo Chamber" is not completely new, but the effects of the issue in the modern world, particularly in the recommendation system, have caused major issues. Echo chamber is a situation where certain ideas, beliefs, and data are developed and reinforced repeatedly in a system, which restricts the opposing ideas and data. In the real world, this echo chamber issue makes people only interact with ideas that only support their existing beliefs, avoiding the opposing ideas[7]. This issue is very obvious in the movie recommendation system, where the user gets recommended with the films that are very similar to the contents that they have already watched, which restricts their exposure to the new genre and diverse contents. In the movie recommendation system, the algorithms that only consider user history and preferences to make recommendations are the cause for the echo chamber issue. The main idea of these algorithms is to provide personalised recommendations that aligns with the user's taste, but by recommending similar contents unintentionally they create the echo chamber

issue[8]. As the system keeps on recommending similar contents, the user will watch the contents that they are already exposed to, which will restrict them to a small selection of movie genres and limits their exposure to various contents[9]. Same as social media echo chamber reinforces political or social biases, movie recommendation systems will reinforce entertainment biases. If a user is watching movies only from a particular country or culture, the user will have a biased understanding of world cinema[7]. From a business perspective, because of this issue if a user gets recommendations only for a particular content, they will miss out the paid premium content which might reduce the revenue of the streaming platform. As an effect of this issue, filmmakers and creators are working on creating content within the popular genres, as those were the ones highly recommended by the system. This will affect the creativity and the innovation of the film industry[9].

1.2 Aim

The main aim of the project is to address and mitigate the echo chamber issue in the movie recommendation system. The echo chamber is the issue where the users are repeatedly exposed to the contents of similar types. This issue can limit the user's exposure to wide range of content. In movie recommendation system this issue make users watch the similar content again and again by recommending the movies with similar genres and qualities. By focusing on this issue, the main goal of the project is to improve the quality and diversity in the recommendations, to make the user get exposed to a wide range of movies. Mitigating this issue not only enhances the user's movie watching experiences, but also promotes a more improved understanding of the world cinema. The motivation of this project was inspired by the increasing need for the recommendation system in the entertainment industry. As the increasing number of users depending on the recommendation system to find new content, the project aim is to provide diverse and unbiased recommendations to the users.

1.3 Objective

The primary goal of the project is to analyse the challenges and the effects of echo chamber issue in movie recommendation systems. The objectives of the project are listed below to address the echo chamber issue in real-world perspective :

1. Understanding the limitations of similar recommendations : The first motive is to identify the challenges of recommending users with the repeated types of similar contents. The goal is to understand how recommendations can limit the cultural exposure, movie viewing experience of the user.
2. Ensuring Balanced User Exposure : In recommendation systems, finding the balance between satisfying user's known preferences and introducing them to new contents is important. The

objective is to provide user with contents that are relevant to their previous behaviour, while also recommending new contents that they might like.

3. Integration of own logic in the existing system : the objective is to integrate the custom logics designed to ensure diversity in the recommendations with the existing recommendation systems.
4. Providing continuous learning and adaptability : As the user preferences, cultures and nature of films getting evolved regularly, the motive is to continuously learn the changing patterns and user feedback.

2 Literature Review

In recent years, there have been major improvements made in movie recommendation systems with the motive of improving user experiences. A movie recommendation system is a technology which automates the age-old idea of word-of-mouth concept. Movie recommendation systems use the data collected from the users with similar preferences or from the similar contents to enhance the viewing experience of the users. As there is development in the system, challenges too will arise. One of the main challenges in the movie recommendation system is the echo chamber issue. The literature review covers the complexity of movie recommendation systems, including both its advantages and disadvantages. The literature review mainly focuses on the echo chamber issue and covers the proposed methods to overcome the echo chamber issue.

2.1 Filtering Methods in Movie Recommendation Systems

The movie recommendation system uses two filtering methods: collaborative filtering and content-based filtering. Collaborative filtering is an effective method in movie recommendation systems. The research paper " Movie Recommender System Using Collaborative Filtering " authored by Meenu Gupta and others, gives an idea of how collaborative filtering is used in movie recommendation systems. Collaborative filtering is a popular method that relies on user interactions. It filters contents based on the similarity of the user. This method predicts the user's preferences by using the user behaviour data. Then the predicted data are compared with the interactions of the other user. Basically, it recommends contents to the user based on the preferences of other users with similar tastes. Collaborative filtering is divided into two types: user-based and item-based filtering. The user-based filtering method predicts the items to be recommended by comparing the data with the other user who has similar interests. The basic concept is that users who have shown similar preferences in the past will have similar tastes in the future. In this method, the system tries to find the similar users to the target user and recommends the contents that the similar users have already liked. The item-based filtering predicts the items to be recommended by comparing the product with similar kinds of products that the user has already shown

interest in. Using similarity measures between the classes, the authors used K-Nearest Neighbours(KNN) algorithm to classify the unlabelled classes. The authors have also used a cosine similarity approach to calculate the distance between the predicted movies and the other movies in the database. The system proposed by the author mainly uses the item-based collaborative filtering, mainly to focus on increasing the overall accuracy and efficiency of the system. At the end, the system uses a hybrid method by combining both collaborative and content-based filtering to enhance the predictiveness of the system[10].

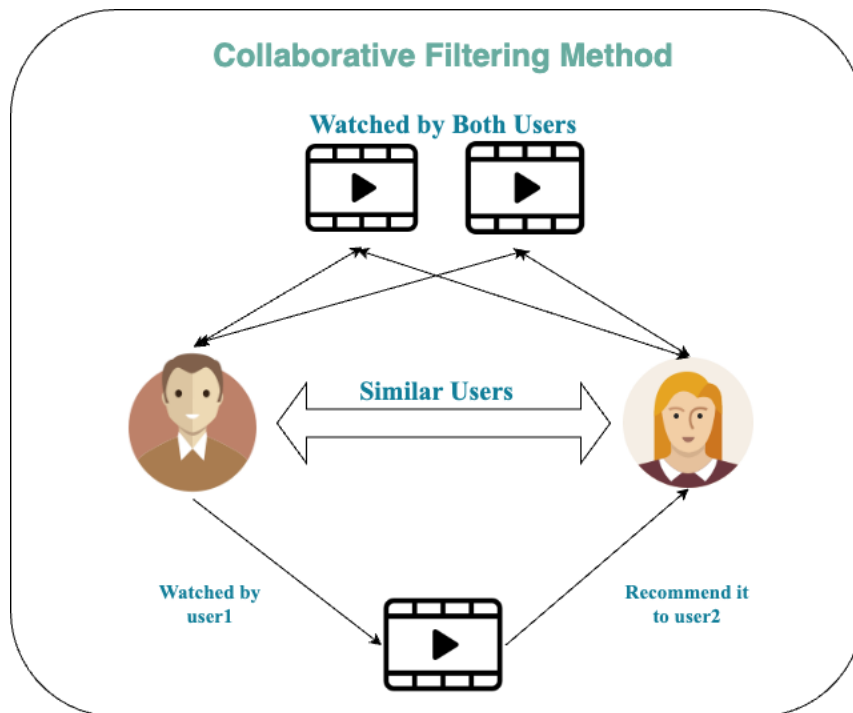


Figure 2.1 Collaborative Filtering Method

The use of Content-based Filtering is popular in recommendation systems, and the research paper titled “Content-based Music Recommendation System” by Aldiyar Niyazov, and others focuses on how it is used specifically within the music recommendation systems and its principles can be adapted and implemented in movie recommendation systems as well. In comparison with collaborative filtering, content-based filtering (CBF) analyses the similarity between items by considering the unique features of the objects itself. It focuses only on the features of the items and does not consider user ratings. The content-based filtering method uses the features of musical compositions to predict the music recommendations. We can implement the same method in the movie recommendation system by using the features of the movies like genre, cast and crew, directors, and plots to predict the movie recommendations. The authors have used Artificial Neural Networks (ANN) to find the similarity between items[11]. The content-based filtering and item-based collaborative filtering might sound similar, but both are two distinct methods. The content-based filtering recommends items by comparing the features of the items, whereas the item-based collaborative filtering recommends items based on

their similarity to the items that the user have already liked in the past. The content-based filtering only requires the feature data of the item. But, the item-based collaborative filtering requires a user-item interaction matrix, to recommend the items.

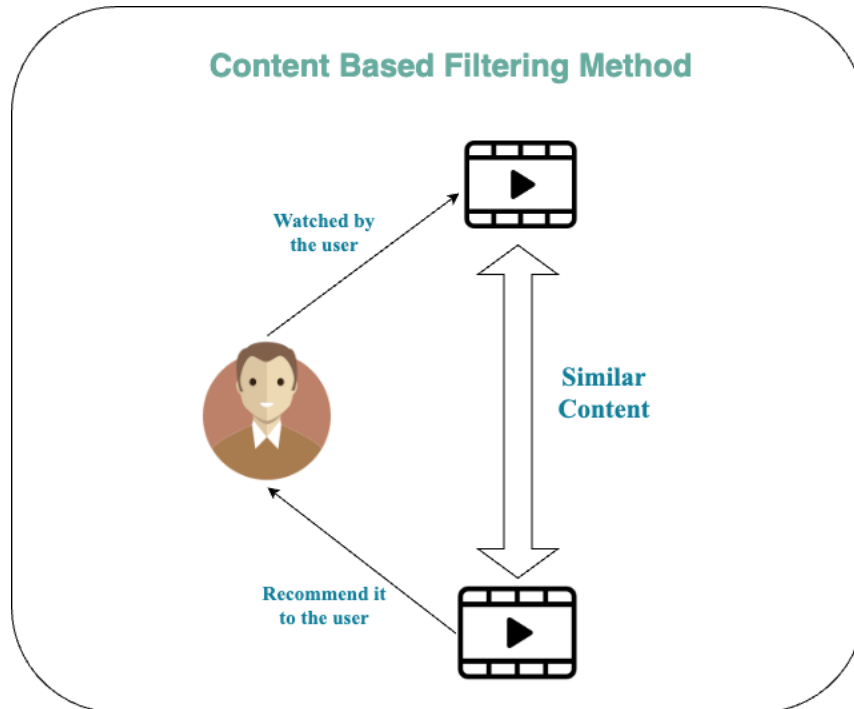


Figure 2.2 Content Based Filtering Method

The research by Meenu et al also gives us the comparison between the collaborative filtering and content-based filtering depending, by highlighting the ability of collaborative filtering method to predict the recommendations by analysing user's behaviour and then comparing it with other users. Whereas the content-based filtering depends only on features of the item to identify related movies [10].

2.2 Hybrid Recommendation System for Movie Recommendations

Hybrid recommendation systems use multiple recommendation methods to improve the overall accuracy and efficiency of the system. The research paper titled "A Hybrid Approach for Movie Recommendation Based on User Behaviour" by Minal V. Jamnekar, and Prof. Sneha U. Bohra gives us an idea of the hybrid system for movie suggestions. The hybrid approach combines both content-based and collaborative-based filtering techniques to offer personalised recommendations which address the limitations of specific recommendation algorithms[16]. The paper proposes a system that uses a dataset which has numerous movie data. This system provides real-time adaptability by using user behaviour. The recommendations are provided by comparing the features of the items and the user profiles. The system includes sentiment analysis techniques that uses movie reviews and gives a success indicator to predict the success rate. The hybrid approach uses the common attributes such as actors,

directors, or writers, and it has the ability to add more attributes to enhance the accuracy of the suggestion[16]. The proposed system uses a user-centric approach, focusing on user choice, interest, preferences, and behaviour. The main goal is to reduce system runtime and enhance accuracy in identifying item relations. In this research an innovative and effective hybrid method is proposed. The proposed method improves from the existing system by using metadata of user behaviour and movies. The research also defines the possible scope of the hybrid recommendation system, suggesting improvements by clustering and equivalence.

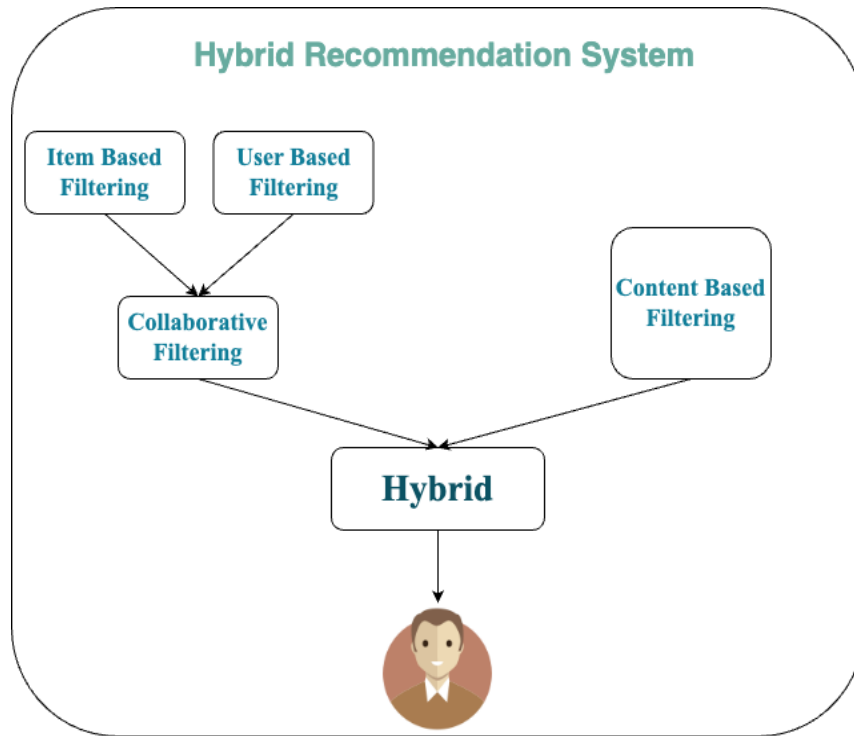


Figure 2.3 Hybrid Recommendation System

2.3 Real World Applications of Recommendation System

2.3.1 Netflix

Netflix has always been at the front of using advanced recommendation systems to increase consumer satisfaction. In a research paper titled "The Netflix Recommender System: Algorithms, Business Value, and Innovation," authored by Carlos A. Gomez-Urbe and Neil Hunt, a detailed study is done on the complex algorithms of the Netflix recommender system[18]. The Netflix recommendation system uses multiple algorithms, each designed to perform a specific task. The main goal of the system is to help users to find movies that are related with their preferences, making sure a personalised viewing experience. The most important part of the recommendation process involves quickly iterating through offline experiments, where historical data is analysed to assess the prediction accuracy of an algorithm[18]. After getting the results from the offline experiments, A/B testing will be conducted. Based on the results of A/B testing the algorithm will be either integrated to the system or will be

modified for further testing[18]. One of the issues that Netflix faces is “Page Construction”. By focusing on a wide range of moods, needs, and situations the organisation can improve the recommendation page. The one more issue is “Member Starting”, where the system finds it hard to recommend contents to a new user. To rectify this issue Netflix have generated a survey while signing up, where new users will have to select contents. From which the recommendation system collects the data and produces the results[18].

2.3.2 Amazon

Amazon is one of the leaders for personalization and recommendations. Their aim is creating unique experiences for the user. The research paper titled “ Two Decades of Recommender system at Amazon.com” by Brent Smith and Greg Linden gives us a detailed study of the evolution and complexities in Amazon's recommendation system[19]. Same as Netflix's recommendation system, Amazon’s recommendation system is also not a single algorithm system. It consists of multiple algorithms each assigned with a different task. The main goal of all these algorithms is to recommend users with the items that they may not know, based on their previous behaviours[19]. At first, item-based collaborative filtering was used to recommend products, which finds out the related products. The algorithm finds out the related products by selecting the products which usually got sold together[19]. The main advantage of this algorithm is its scalability and the real time adaptability. But the main disadvantage of this algorithm is that by finding out the related items by taking together products into consideration the system will create a bubble where similar kinds of products will get recommended with the same products. This will reduce the impact of recommending more diverse products to the customer.

2.3.3 YouTube

YouTube is also one of the most popular online video platforms, and has been a focus for content exploration and recommendations. The research paper titled “The YouTube Video Recommendation System” by James Davidson et al. gives us the information of how the system has evolved and its challenges[20]. With the previous behaviours of the user, the recommendation system has been built to recommend personalised contents. The system also focuses on suggesting new, diverse, and relevant contents to the user that they might like. The main method used in the YouTube recommendation system is rule mining or co-visiting counts. By taking user watch sessions into consideration this methodology calculates the frequency at which pairs of videos are co-watched during each session[20]. The relatedness score is then calculated based on the co-visit counts of one video to the other video. Then based on how popular the video is, the relatedness score is normalised[20]. But the main challenge in this method is that it does not necessarily mean that a user enjoyed a video because they have watched

it. So, if they haven't enjoyed the video the system will keep on recommending similar contents, which then creates a bubble.

2.3.4 Kindle

The book industry has seen a major change due to the development of digital platforms like Amazon Kindle. This leads to the important change in the way of how readers are going to consume with the books. With the increase in the e-books, there has been an increase in the need of recommendation system as well. The research paper titled "What books will be your bestseller? A machine learning approach with Amazon Kindle" authored by Lee et al. gives us the methodology of how the Amazon Kindle recommendation system works. The research highlights the importance of online bookstores, where most users buy their books. Like many of the Amazon's recommendation system, the Kindle recommendation system also contains multiple algorithms to recommend books to the readers. The recommendation of kindle focuses on similar items, similar users, purchase history, browsing history and ratings to recommend the books. The traditional recommendation systems mainly use the collaborative and content-based filtering methods, which leads to issues like echo chamber and cold start problems. Lee et al. proposed a solution to these issues by analysing whether the user satisfaction can be predicted using available book descriptions. The research paper also finds few limitations in the existing system. The system can face inconsistencies in predictions, as there is a gap between the descriptions on the main page of a book and the full content. In order to attract readers, the authors can add descriptions. The paper also proposes different perspectives on how the book recommendation system will develop in the future[24].

2.4 Related Works

2.4.1 Echo Chamber and Filter Bubble Issue in Recommender Systems

The research paper titled "Filter bubbles in recommender systems: Fact or fallacy—A systematic review" authored by Qazi Mohammad Areeb and others, focuses mainly on the echo chamber and filter bubble issue in recommendation systems. The echo chamber issue and filter bubble issue has become the major issue, because of the increased use of recommendation systems in recent times. Echo chamber is an issue where users get recommended with similar types of contents which restricts their exposure to diverse contents. In movie recommendation systems, this issue can result in a reduction in content diversity, as users are mainly suggested with movies that are related to their previous viewing patterns. In business industries, the filter bubble effect creates an increase of the "Matthew effect" between important organisations[12]. This impact develops as a tendency to recommend popular items, while ignoring products and information that leave from the long tail hypothesis. Since users are biased towards popular products, it leads to decrease the sales of diverse products and might constrain the overall profitability. In addition, filter bubbles can indirectly promote the increase of unwanted content

on online social media platforms, including the spread of rumours and rumours[12]. The existing systems can create the filter bubbles because their main focus is on improving the accuracy of recommendation rather than recommending diverse products. To avoid the filter, bubble the author have suggested to include random recommendations to increase the diversity. The proposed method mainly focuses on finding the balance between personalized and diverse recommendation. Both the personalized and diverse recommendations are important but incorporating them is the issue. To avoid this issue, the author has suggested a multi-objective optimization problem. This approach tries to recommend multiple sets with different values of personalization and diversity. With those different set of recommendations, the proposed method tried to find a balance between them, resulting in personalized recommendations while also being diverse[12].

2.4.2 Understanding Echo Chambers

The research paper titled “Towards a Standardised Methodology for Identifying Echo Chambers: A Reddit Case Study” authored by Virginia Morini, Laura Pollacci, and Giulio Rossetti provides a detailed exploration of the echo chamber issue on the Reddit platform[21].

The concept of echo chambers, where users are exposed to contents that reinforces repeated contents within a closed system, isolated from different contents. This issue can restrict the democratic flow of different contents. The authors proposed a framework to identify the echo chambers with common features shared by online platforms. The proposed framework consists of four-stage pipelines. The first one is “Controversial Issue Identification”, where it recognises the emergence of opinion polarisation comes from topics that has major differences. The second one is “Users' Ideology Inference”, after identifying the controversial issue, the second step is the process of determining the user’s likelihood toward the controversy. It is identified by analysing the user’s past behaviours. The third one is “Users Debate Network Definition”, where user’s interactions are analysed and labelled based on their previous behaviour. The final one is “Homogeneous Meso-scale Users’ Clusters Identification”, a method to identify network areas which shows similarity. After defining the labelled interaction networks, the paper uses a group detection algorithm “Eva”, which has been designed to detect groups that optimise their label similarity. The necessity to separate the entire network into clusters that shows similarity was discussed by the authors. The goal is to optimise both topological clustering and the label similarity in all communities[21].

2.4.3 Insights from User behaviour Analysis

In recommendation systems the echo chamber is an important issue as it reinforces the user preferences by algorithmic recommendations and restricts their exposure to diverse contents. In movie recommendation systems, the mentioned issue gets more importance due to the potential limitation of user’s cinematic experiences resulting from continuous exposure to similar genres and contents[13].

The research work conducted by Yingqiang Ge et al. titled "Understanding Echo Chambers in E-commerce Recommender Systems" provides methods for overcoming the echo chamber issue in movie recommendation. To mitigate the echo chamber, issue the author proposed a method, which splits the user behaviours into separate blocks, including browsing, clicking, and purchasing activities. This splitting method is used to understand the user's behaviour and preferences and by using this information we can modify the algorithms to increase the diversity in recommendation. The author has proposed a clustering method to modify the algorithm to increase diversity. The method clusters the items based on their features and then the system will recommend one item from each cluster to ensure diversity. Also, the author proposed the hybrid model as a possible solution to increase the diversity. User classification, as studied by Ge et al., holds important value in movie recommendation systems. Categorising users into Following and Ignoring Groups based on their response to recommendations is an approach to promote content diversity. This, in the end, helps to mitigate the echo chamber effect [13]. Then by analysing the reinforcement patterns, the echo chamber issue can be resolved. This research by Ge et al provides different frameworks like user behaviour analysis, reinforcing patterns, and classification, which helps us to overcome the echo chamber issue in movie recommendation systems.

2.4.4 Fairness-Aware Approach

The echo chamber issue in the movie recommendation system is created by the presence of biased link predictions in network analysis. The research paper titled "Bursting the Filter Bubble: Fairness-Aware Network Link Prediction," authored by Farzan Masrour and others, provides valuable methods to solve the echo chamber issue in movie recommendation systems[14]. The filter bubble issue in network link prediction, as studied by Masrour et al., shows similarity with echo chambers in movie recommendations. The correlation between the protected variables, such as gender, and link predictions adheres with users' movie preferences resulting in limited exposure to specific content[14]. The author has used three datasets : Facebook ego network, Google+ and, Dutch school network. In this study, Masrour et al. proposed a set of fairness criteria, including demographic parity and equalised odds, that can be directly applied to movie recommendations. The demographic parity considers the degree of independence between the model's output and the protected attribute. The equalised odds make sures that the predictions are independent of the protected attribute[14]. By using these criteria, we can avoid the echo chambers and recommend more diverse contents to the users. In the same way, by using the subgroup dyadic-level protection, we can prevent demographic-specific concentration, which helps in recommending diverse content and mitigating echo chambers[14]. The subgroup dyadic-level protection ensures that the model's predictions are not biased against specific subgroups. This fairness criteria can be defined by considering the protected group membership of the individuals presented in the links[14]. In movie recommendation, promoting cross genre or cross theme suggestions increases

the diversity of content exposure. This approach overcomes the echo chamber issue. By analysing user behaviour and preferences, algorithms can be built to increase content diversity[14]. The author has used accuracy and the ROC curve as the evaluation metric. The research paper summarises that using the fairness-aware method we can build a more effective movie recommendation system and avoid the echo chamber issue.

2.4.5 Addressing the Echo Chamber through Polarisation Detection

In recommendation systems, the echo chamber issue and group polarisation create significant problems. In the publication titled "An Improved Recommendation Algorithm for Polarised Population" Wang, Zhao, Huang, Dai, and Zhu presents new methodologies which provide effective solutions for mitigating the echo chamber issue in movie recommendation systems[15]. This paper highlights the importance of collaborative filtering algorithms in personalised recommendation systems, especially in the context of social media platforms. These algorithms use user behaviours, such as likes and interactions, to provide personalised recommendations. But the major issue is the unintentional creation of the “passive” group polarisation. This issue makes the users to receive similar recommendations which results in the echo chamber effect. This paper presents a methodology to address the recommendations for the polarised users. By using a combination of user behaviour and feature information, the algorithm assigns a weight to different behaviours. To determine if the user is in polarised group or not, the author used a polarization score. The polarisation score is calculated by assigning weights for users with high or low viewing volumes. Also, the proposed algorithm normalises the score to avoid the deviation in the score for user with extreme behaviours. For the users identified in the polarised groups, the algorithm uses word2vec method to process the data and K-means clustering to find and remove the data that contributes to the polarisation. The use of this method in a movie recommendation system helps in mitigating the reinforcement of existing biases. The authors have used two real world which highlights the relevance of the proposed algorithm. The results, as claimed in the paper, shows that the proposed algorithm can be adapted to movie recommendation system by incorporating it with the collaborative algorithms. This suggests that this algorithm will help in mitigating the issue of crowd polarisation and echo chambers.

2.4.6 Addressing the Polarisation

The research paper titled “Controlling Polarisation in Personalization : An Algorithmic Framework” authored by L. Elisa Celis, Sayash Kapoor, Farnood Salehi, and Nisheeth Vishnoi discusses the challenges of personalization in online platforms, with a focus on polarisation and the echo chambers. The authors proposes that the existing personalisation and recommendation systems enhances the accuracy for both the users and the platforms, but it also creates biases that impacts the results[22]. The authors also discussed about the filter bubble issue where the system creates a bubble where users get

stuck with their own cultural or ideological bubbles, without the exposure to the diverse content[22]. To overcome this issue the authors have proposed a framework, which allows users to set limits on the distribution from the selected content. By allowing the users to set limits on the categories or genres of the content they want, they have overcome the echo chamber issue. As the user set the limits on the content types, the system continues to learn new patterns and personalise the content according to it[22]. The proposed framework is evaluated on an online news article dataset. The results were also provided proving the ability of the system to control polarisation[22]. Additionally, they have also tested the algorithm on MovieLens dataset. The result from this test shows the adaptability of their approach to movie recommendation systems by highlighting the algorithm's ability to recommend diverse movies and not getting stuck in the echo chamber[22].

2.4.7 Breaking the Echo Chamber by “Cousins” Method

The research paper titled “Escaping the Echo Chamber in Movies Recommender System” by Anthony El Haddad addresses the issue of individuals getting stuck in their own cultural and ideological bubbles, known as the "echo chamber" or "filter bubble" issue. The echo chamber issue is described as a situation where the user is continually exposed to the movies that are related with their past preferences which leads to the repetitive content and restricts the diverse content. In this research paper the proposal of an innovative method ensures a seamless process of recommendations. It has been found that users who regularly watch movies from a particular genre continue to receive similar content recommendations from the system[17]. To solve this issue, this paper presents a proposed approach for escaping the loop created by recommendation systems. The main purpose of this research is its proposal for successfully handling the echo chamber issue, where a recommendation is provided, that is related to the current film but slightly different, this is known as "cousins"[17]. The word “Cousins” shows that these movies share a very tight relationship with the user’s preference but also, they introduce an element of diversity. The method suggested in this paper is expected to perform well across various user profiles and applications.

2.4.8 Regularizing Score to avoid Popularity Bias

The main method by which the recommendation systems gain information about the user preferences is through indirect inputs, which includes actions such as clicks, purchases, and item consumptions. But an issue these systems faces is the popularity bias. The paper titled “ Countering Popularity Bias by Regularizing Score Differences" by Wondo Rhee, Sung Min Cho, and Bongwon Suh, suggests us a method to overcome the bias. When the data is trained for recommendation system, the recommendation system tend to assign a high recommendation score to the popular items. This issue is known as model bias. Because of this issue the system is likely to develop a situation where the users get stuck in the echo chamber, which results in recommending the same popular items and misses out the diverse items.

This results in less personalised recommendations. It not only reduces the range of diverse content the users are exposed to, it also reduces the personalization. The authors propose that by regularizing the score of recommendations, the system ensures the consistency among the items preferred by the user. The authors introduced the regularisation term by using the pairwise loss, which optimises the score between the preferred and unpreferred items. By regularising the score, the system reduces the score differences between these items, which results in high debias and high accuracy performance. The authors have evaluated their proposed method's effectiveness by experimenting on a synthetic dataset, which includes model bias with baseline training. The results given in the paper shows that the model bias is reduced with maintaining a high accuracy. When comparing this proposed method with the existing systems, it showed advantages in terms of efficiency[23].

3 Methodology

3.1 Content Based Filtering

Content-based filtering is a method used in recommendation systems to recommend movies to the user based on their previous behaviour and preferences. The content-based filtering method compares the features of the movies with the movie that the user has already watched[25]. In content-based filtering, movies are represented by their features. By giving an input of movie title, the model finds movies with similar features and recommends them to the user. The K-Nearest Neighbours model is used to find the movies with the most similar features of the input movie.

3.1.1 K-Nearest Neighbours

The K-Nearest neighbours(K-NN) algorithm is a type of instance-based learning algorithm, which means it does not actually learn a model. But instead, it stores the instances in the memory and generates predictions based on how similar the new instance and the stored instance is. The main aim of the model is to calculate the distance between the new instance and the stored instance. There are many ways to calculate the distance between the instances. One of the most popular and effective methods is cosine similarity. The 'K' in the K-NN is used to represent the number of nearest neighbours to consider when predicting. A smaller K value can be noisy and sensitive to outliers, the large value of K can ease the decision boundaries, but the large value will include data from other classes[26]. K-Nearest neighbours is used for the content-based filtering as it makes decisions based on the entire dataset, where other algorithms will require to be trained to make predictions. In the real-world applications, the data of the user and movies keeps on updating so K-Nearest neighbours would be very helpful as it does not require a training phase.

3.1.2 Cosine Similarity

Cosine Similarity is a popular metric used in the recommendation systems, specifically when it comes to finding the similarity between two items. The cosine similarity is used in the system to measure the angle of similarity between the movies. The core part of the cosine similarity is the cosine measurement of the angle between two vectors. The value of cosine similarity ranges from -1 to 1[27]. Here vectors acts as the movies being compared with each other. Cosine similarity is used in the system as unlike other metrics, it only focuses on the angle between two vectors. Since the dataset is high dimensional, cosine is chosen as a metric, as it was not concerned about the dataset's size or length. It makes the computationally efficient as it only considers the non-zero dimensions.

$$\text{cosine similarity} = \frac{A \cdot B}{||A|| \times ||B||}$$

Where A and B are the two vectors being compared. The $A \cdot B$ is the dot product of the vectors and the $||A||$ and $||B||$ are the magnitudes of the vectors. If the angle between the two vectors is 0° , the value of the cosine will be 1, which indicated that the vectors are identical and similar to each other. If the angle between the two vectors is 90° , the cosine value will be 0, which means that the vectors are dissimilar. If the angle between the vectors is 180° , the value of cosine will be -1, which indicates that both the vectors are in opposite directions[27].

3.2 Collaborative Filtering

The content-based filtering recommends movies only based on the similarity of the users. But the collaborative filtering recommendations are made using the data collected from the other users with the similar preferences like the test user. In the built system the collaborative filtering makes recommendations based on two methods : user-user similarity matrix and user-item matrix. The user-user similarity matrix is calculated using the cosine distance. Where the users are the vectors to calculate the distance between them.

3.2.1 User-User Matrix

User-User matrix works on the idea that users who have already shown interest to a particular content are more likely to show interest again in the future. The user-user matrix uses the cosine distance to find the users who are similar to the target user, with the similarity matrix the system recommends the contents to the target user that the similar user have already consumed. With the system found the similar users to the target user, then the system will find the contents that the similar users have already consumed using the user-item matrix.

3.2.2 User-Item Matrix

Item-Item matrix uses the user ratings of the content and then find the items with the similar ratings. If two users rates the contents as the same, they are considered as similar users. Same as the users, if two items gets rated as the same by different users, they are considered to be similar. After the system find out the similar items and users using user-user matrix and item-item matrix, the system recommends the similar items to the target user that the similar user have already consumed.

3.3 Singular Value Decomposition

Singular Value Decomposition(SVD) is a mathematical method that decomposes a matrix into three other matrices. The SVD factorizes the original matrix into three other matrices, this factorization, retains all the main structures of the data. The SVD retains only the largest singular values and their corresponding singular vectors to obtain the low-rank approximation of the original matrix. The noise and the less important features are removed as these approximation captures the most important patterns in the data.

$$A = U \Sigma V^T$$

Where A is the original matrix, U is the matrix representing the left singular vectors of A, Σ is a diagonal matrix which contains the singular values of A, and V^T is the transpose of the matrix representing the right singular vectors of A. The singular values in Σ represent the strength of the corresponding singular vectors. The large singular value shows that their corresponding singular vectors are more important. In the recommendation system, the user-item interaction is represented by the matrix A. The SVD model often produce better results than the usual collaborative filtering model. As the data gets increased every day, considering the large dimensionality of the data, the SVD method is used in the system as SVD allows for dimensionality reduction without losing the most important patterns[28].

3.4 Hybrid Model

The Hybrid model is a most effective method for the recommendation systems as it combines the strengths multiple recommendation models. The main objective of the model is to enhance the quality of recommendation by overcoming the challenges in each model with the strengths of one other. In our system the hybrid model combines both the content based and collaborative filtering methods. The content-based model recommends contents only based on the features of the item which might result in creating the echo chamber issue. The collaborative filtering model recommends item based on the preferences of the similar users which might result in dissimilar recommendations. The hybrid model tries to address the drawbacks and build on the advantages of each model separately by combing them. This will result in recommending more accurate. Furthermore, the hybrid model in our system includes three more methods to ensure that the system does not creates the echo chamber issue.

3.4.1 Serendipity

Serendipity refers to the method where the unexpected recommendations are included in the recommendation lists. As the recommendation system focuses on recommending the similar contents to the user, serendipity ensures recommending the unexpected items that the user have not exposed to. These recommendations are not only based on the user's behaviour, but also it includes an element of novelty and surprise[29]. Serendipity in our system is all about introducing the user to possible new contents, which helps in overcoming the echo chamber issue. In our system a certain percentage of recommended movies are replaced with the movies that are not in the recommendations to make sure that the final set of recommendations is a mix of similar contents with few unexpected movies to make sure that the system is not creating an echo chamber.

3.4.2 Diversity

Even though the main motive of the recommendation systems is to provide personalized recommendations, it is also important to maintain diversity in the recommendations. Diversity ensures that the recommended items are not just relevant but also varied, providing a wide range of user preferences[30]. The main goal of introducing diversity in the system is to make sure that the system is not providing very narrow or biased recommendations, which results in the creation of echo chamber issue. There are many ways to ensure diversity. In our system the diversity is ensured by monitoring the genres of the recommended movies. The diversity factor makes sure that the recommendation does not have movies only with a particular genre. This ensures a balanced and varied set of recommendations.

3.4.3 Maximal Marginal Relevance

Maximum Marginal Relevance(MMR) is a method used in information retrieval and recommender systems to diversify the set of recommendations. To make sure the user receives the wide range of recommendations, this method tries to find a balance between the relevance and diversity in the recommendations set. The MMR algorithm continuously selects the items that maximizes a specific criterion. This criterion is a combination of the item's similarity to the user's query and its dissimilarity to the already selected items. The balance between the similarity and diversity is controlled by a parameter β .

$$MMR = \arg \max_{d \in R/S} [\beta \times \text{Relevance}(d, q) - (1 - \beta) \times \max_{d' \in S} \text{Similarity}(d, d')]$$

Where R is the set of all items, S is the set of selected items, q is the user's query or profile and β is the balance parameter between relevance and diversity[31].

4 Implementation

4.1 Dataset

For mitigating the echo chamber issue in the movie recommendation at first, we've to build a basic movie recommendation system. For building a recommender system the MovieLens 25 million dataset is chosen. This dataset is often known as ML-25M dataset. ML-25M is one of the largest and most used datasets for building a movie recommendation system. This dataset is provided by the GroupLens research team from the University of Minnesota[32]. The ML-25M dataset is a good choice for building the recommendation system as it has numerous movie features and user behaviours. The dataset contains 25 million ratings that 162,000 users have given to 62,000 movies. The recent dataset was released in December 2019. The ratings in the dataset are given in half-star increments on a scale of 0.5 to 5. Also, the dataset contains movie titles, genres, ratings, and user-provided tags that describes different movie features. The ML-25M is selected for this project, as the dataset contains 25 million user ratings which ensures a wide range of user behaviour and preferences are recorded in the system. This wide range of user data is needed to ensure the diversity in the recommendation system.

4.2 Data Pre-Processing

Data pre-processing is the very essential part of our project as it involves cleaning the data, transforming, and preparing the unprocessed data to use in algorithms. The main goal is to enhance the quality of data to make it suitable for further analysis and ensure that the models built will perform more accurate.

4.2.1 Data Cleaning

To start with, the columns "timestamp" and "title" are removed from the data frames as reducing the dimensionality of the data frame is more important. The timestamp represents the exact time that a user rated a movie. The reason this attribute is dropped is that the exact time a user rated a movie will not add any value to the system as the user's behaviour changes over time and timestamp does not helps us in identifying the user's taste. The title column represents the title of the movie. As the main goal of the recommendation system is to find the movies with similar features like genres rather than finding movies with similar titles, the title column has been dropped. But the title column helped in extracting new feature about the movie, which is discussed in the feature engineering section. So, to ensure that the recommendation system mainly focuses on the features that are important to gather the details of the movie, these columns have been dropped from the data frame.

4.2.2 Feature Engineering

Feature engineering is the process of creating new features with the existing data, that would add value to make the algorithms work more accurately. Feature engineering is very important in the context of our project as it would help the algorithms to enhance the quality of the recommendations.

4.2.2.1 Extraction of Year

Even though the column “title” was planned to be dropped, before dropping the column the release year of the movie was extracted from the movie titles. The creation of the column “year” provides the important detail of the period the movie was released. This helps the algorithm to provide more personalised recommendations as it will consider recommending movies from the same period. The quality of the movie changes over time as the technology of the film making evolves over time, as a result the year feature helps in recommending movies to diverse users who has preferences for certain quality of movies.

4.2.2.2 Transformation of Genres

The “genres” column of the data frame has the details of the movie genres. The genre information of the movies is transformed from a concatenated string format into a one-hot encoded format. One-hot encoding is a widely used method to convert the categorical data into a format that used in algorithms. The one-hot encoder uses N state registers with N bits to represent N independent objects[33]. Here for each genre, only one status bit is activated by assigning a value of “1”, whereas the others are set to “0”. This one hot encoding makes the genres represented as a binary column where “1” shows the presence of the genre and “0” shows their absence. Representing genres as binary values makes the algorithms to process it easier.

(no genres listed)	Action	Adventure	Animation	Children	Comedy	Crime	Documentary	Drama	Fantasy	Film- Noir	Horror	IMAX	Musical	Mystery	Romance
0	0	0	1	1	1	1	0	0	0	1	0	0	0	0	0
1	0	0	1	0	1	0	0	0	0	1	0	0	0	0	0
2	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1
3	0	0	0	0	0	1	0	0	1	0	0	0	0	0	1
4	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
62418	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
62419	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
62420	0	0	0	0	0	1	0	0	1	0	0	0	0	0	0
62421	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
62422	0	1	1	0	0	0	0	0	1	0	0	0	0	0	0

62013 rows × 19 columns

Figure 4.1 Encoded Genres

4.2.2.3 Creation of user_movie matrix

For collaborative filtering to find the movies that the user has already watched, the “user_movie” matrix was created. The “user_movie” matrix is the representation of user ratings for the specific movies. Each row of the matrix represents a unique user, and each column of the matrix represents the unique movies. The intersection of the user and movie represents the user rating to that specific movie. If the user has not rated the movie the value will be “0.0”. In our project a subset of 2500 active users and popular movies is made, which helps the algorithm to provide more accurate recommendations. By taking the subset data reduces the computation of the system.

title	'burbs, The (1989)	(500) Days of Summer (2009)	10 Cloverfield Lane (2016)	10 Things I Hate About You (1999)	10,000 BC (2008)	101 Dalmatians (1996)	101 Dalmatians (One Hundred and One Dalmatians) (1961)	12 Angry Men (1957)	12 Years a Slave (2013)	127 Hours (2010)	...	Zero Dark Thirty (2012)	Zero Effect (1998)	Zodiac (2007)	Zombieland (2009)	Zoolander (2001)
userId																
187	0.0	4.0	0.0	3.0	0.0	0.0	0.0	0.0	0.0	4.0	...	0.0	0.0	4.0	4.5	0.0
541	0.0	0.0	0.0	0.0	0.0	0.0	0.0	5.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0
548	0.0	3.5	0.0	4.5	0.0	2.0	4.0	0.0	0.0	3.5	...	0.0	0.0	4.5	4.0	2.0
626	2.5	0.0	0.0	4.0	0.0	4.0	4.0	4.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0
653	0.0	0.0	0.0	4.0	0.0	4.0	5.0	3.5	0.0	0.0	...	1.0	0.0	0.0	0.0	0.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
162047	0.0	4.0	0.0	3.0	0.0	2.5	0.0	5.0	4.0	4.0	...	3.5	3.5	5.0	3.0	2.0
162271	0.0	3.0	2.5	0.0	0.0	1.0	0.0	3.5	3.5	4.0	...	3.5	2.0	4.0	3.5	2.5
162495	3.5	0.0	0.0	2.0	0.0	1.5	1.5	0.0	0.0	0.0	...	0.0	0.0	0.0	5.0	3.5
162508	0.0	2.5	0.0	3.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	5.0	3.0	0.0
162516	0.0	4.0	3.5	3.5	0.0	3.5	4.0	4.5	4.0	4.0	...	4.0	4.0	4.0	3.0	3.0

2500 rows × 2500 columns

Figure 4.2 user_movie matrix

4.2.2.4 Creation of user_similarity_matrix

Same as the “user_movie” matrix, the “user_similarity_matrix” was also created for using in the collaborative filtering algorithm. In the “user_similarity_matrix” both the rows and columns represents the user Ids, where each cell of the matrix has the similarity score between the users. The similarity values in the matrix are calculated by using cosine similarity. The cosine similarity measures the similarity between the movie rating patterns between the users. The “user_similarity_matrix” is calculated only for the subset of 2500 active users. The reason why a subset of active users was taken is, in the whole dataset there are users who have rated only few movies, such users will not provide more information to the algorithms. By using the subset data, the system ensures more accurate recommendations.

userId	187	541	548	626	653	757	803	846	847	901	...	161560	161586	161675	161826
userId															
187	0.000000	0.377109	0.619964	0.328434	0.345707	0.317984	0.244805	0.286042	0.302873	0.523440	...	0.481456	0.395431	0.444307	0.279882
541	0.377109	0.000000	0.399563	0.369528	0.339883	0.325299	0.298574	0.382702	0.486317	0.562154	...	0.552356	0.468059	0.430007	0.392127
548	0.619964	0.399563	0.000000	0.433299	0.373833	0.398693	0.335730	0.413175	0.454961	0.545979	...	0.537080	0.496341	0.638029	0.384246
626	0.328434	0.369528	0.433299	0.000000	0.533515	0.510231	0.364587	0.482578	0.532921	0.292888	...	0.364735	0.558424	0.341678	0.406413
653	0.345707	0.339883	0.373833	0.533515	0.000000	0.374188	0.261879	0.346212	0.382055	0.301018	...	0.335936	0.410705	0.250162	0.311981
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
162047	0.506394	0.537061	0.648291	0.473462	0.377559	0.516850	0.462124	0.558417	0.658392	0.549948	...	0.545291	0.623195	0.573317	0.568303
162271	0.362515	0.468968	0.459307	0.500142	0.413919	0.540284	0.489419	0.494453	0.704761	0.492067	...	0.398446	0.571641	0.410425	0.480960
162495	0.535837	0.354390	0.602432	0.484528	0.426847	0.360092	0.287027	0.366069	0.392697	0.447523	...	0.454687	0.502105	0.488929	0.352741
162508	0.335815	0.359071	0.403801	0.391955	0.267976	0.355265	0.330253	0.347030	0.438468	0.353967	...	0.338637	0.407542	0.378986	0.331248
162516	0.543190	0.528265	0.669688	0.598355	0.499238	0.603234	0.498946	0.560399	0.672512	0.569944	...	0.545499	0.634456	0.558250	0.514525

2500 rows x 2500 columns

Figure 4.3 user_similarity_matrix

4.2.3 Scaling the Genre Features

While using algorithms it is important to scale the features. After the genres of the movies being transformed into the one-hot encoded format, the genres are scaled to make sure all features have the same scale. The genres are scaled using “standard scaler”. The standard scaler standardized the genres features by removing the mean and scaling them to their unit variance. The formula for the standardization is :

$$Z = \frac{x - \mu}{\delta}$$

Where “Z” is the transformed value, “x” is the original value, “μ” is the mean and “δ” is the standard deviation[34]. By scaling the genre features, it improves the performance of the algorithms as the algorithms depends on the distance of the data points.

4.3 Data Integration

For building algorithms of the recommendation system, data integration is very important, as integrating multiple data frames into one removes the need to continuously cross-reference between data frames. The data integration in our system ensures that all the movie details are consistently paired with the other features. The data frames in our system are merged using a common feature of “movieId”. “movieId” is the primary key in the main data frame as it an unique identifier for the movies. Using “movieId” as the merging key make sures that ratings of the movies are paired collected with the respective movies. By ensuring this consistency, the algorithms will easily learn patterns and trends between movie attributes and user preferences.

4.4 Exploratory Data Analysis

Exploratory Data Analysis is the most important step in the project, as it helps to understand the dataset's trends, correlations, and patterns. The information about all the features in the dataset are summarised and visualised, which helps in selecting what models should be used for the project. Exploratory Data Analysis also helps in finding out the important features that has the more impact on the recommendation system.

4.4.1 Data Integrity Check

Ensuring the data integrity is important for building the system, as if the data is inaccurate or inconsistent the recommendations made by the algorithms will also be inaccurate. Missing values in the data will impact the performance of the algorithms. So, to ensure that all the data is consistent, the data is checked if it contains any missing values, the data types of all the features in the data is checked and the statistics of the data is also checked to get more information about the features.

4.4.2 Distribution of Ratings

The distribution of ratings is an important pattern to understand the user behaviour and the evolution of the recommender system. To understand how the user rating behaviours have evolved, visualised the trend of ratings over time. By visualizing this we can find the years where the user activity has increased or decreased. By grouping the number of ratings per year the data is visualized. The following figure shows the visualization of distribution of ratings over time.

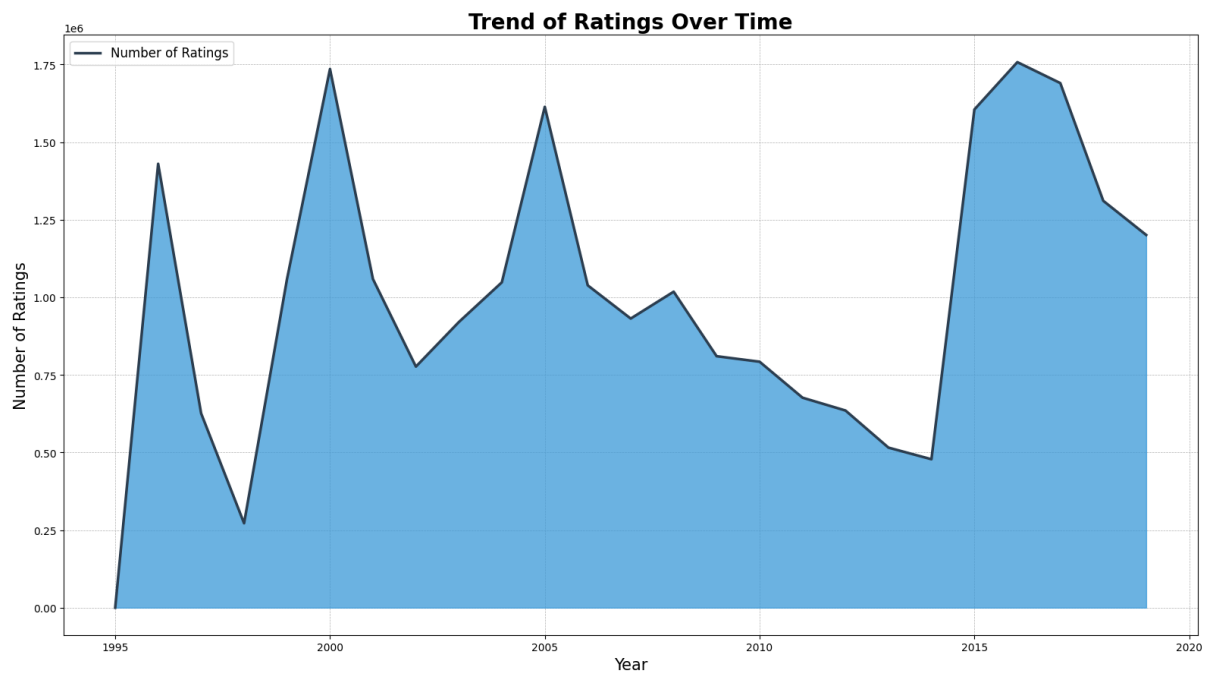


Figure 4.4 Trend of Ratings Over Time

4.4.3 Distribution of Movies Across Genres

The distribution of movies across genres helps in understanding the genre distribution in the dataset, which also helps in avoiding the bias towards overrepresented genres. The distribution of movies across genres is visualized using a bar plot. By being aware of the genre distribution, the system can balance the recommendations to ensure diversity in the recommendations. The following figure shows the bar plot of distribution of movies across genres.

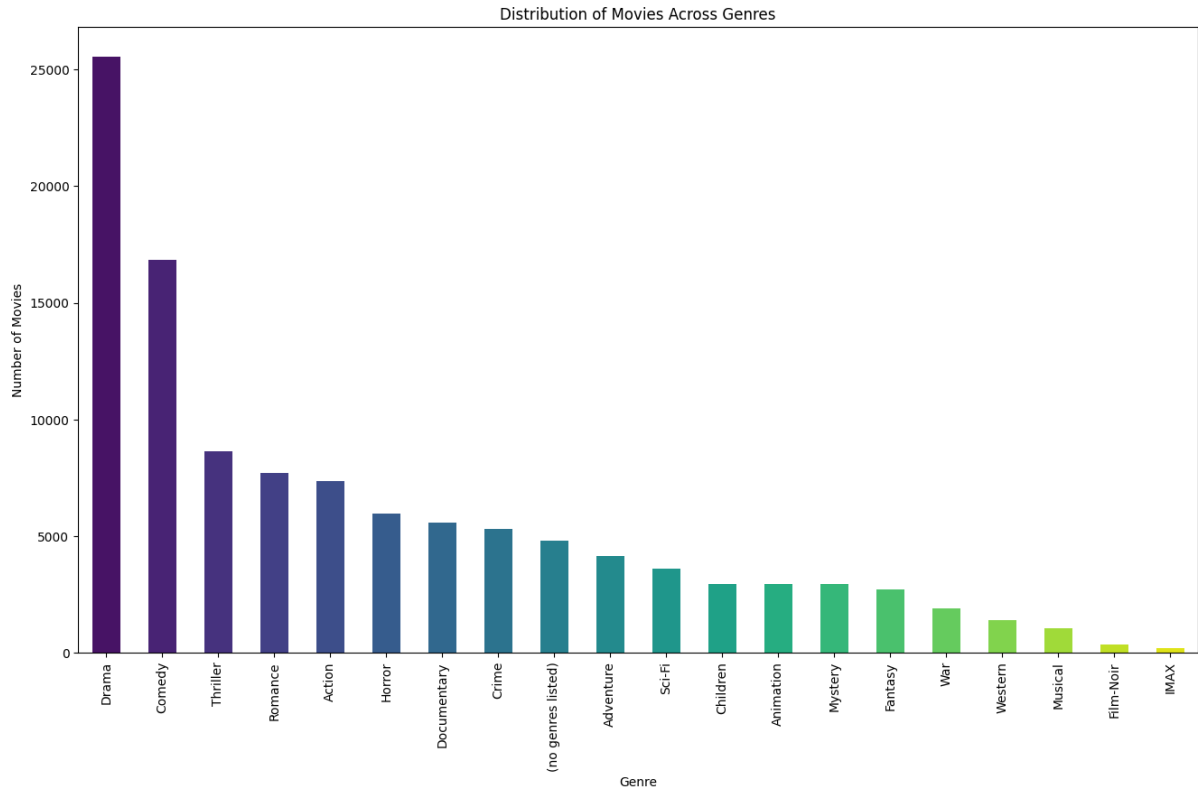


Figure 4.5 Distribution of Movies Across Genres

4.4.4 Year-Wise distribution of Movies

The distribution was visualized to analyse the year-wise distribution of movies, by which we can identify the specific period where the movie production was high or low. The following figure shows the distribution of movies over years.

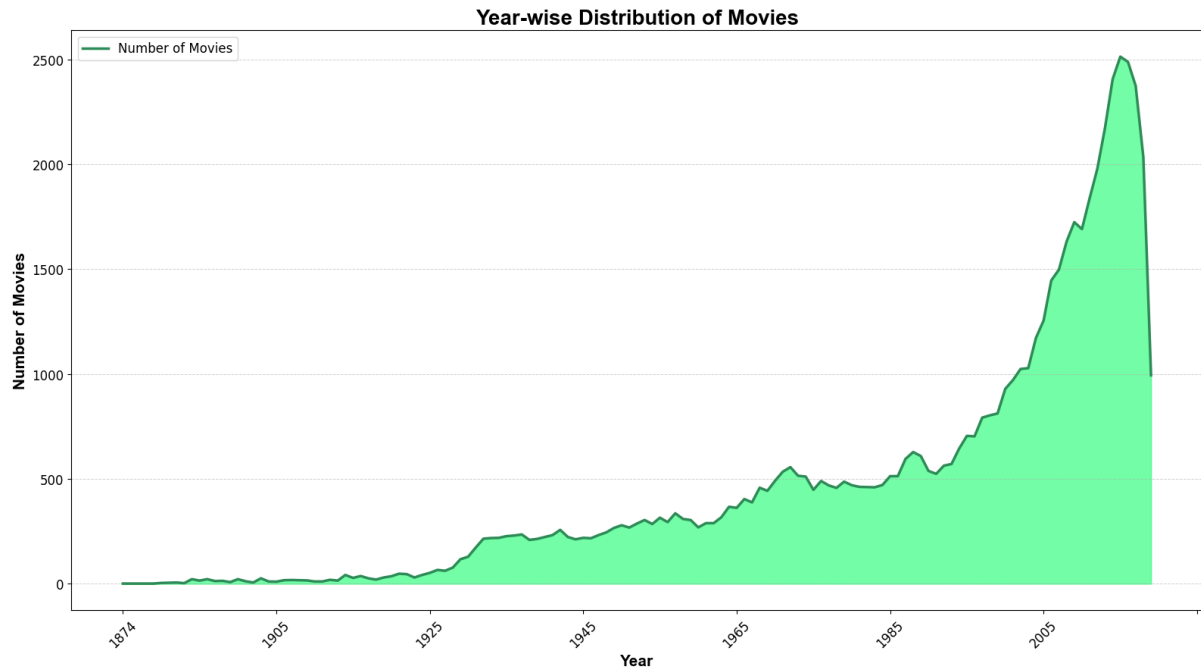


Figure 4.6 Year-Wise Distribution of Movies

4.4.5 Genre specific Ratings Analysis

The genre of a movie is important in determining its reach to different users. By analysing the average ratings across genres, we can understand that which genre is more liked by the users, and which are less liked by the users. The rating across genres is visualised using a bar plot. The following figure shows the rating of each genre.

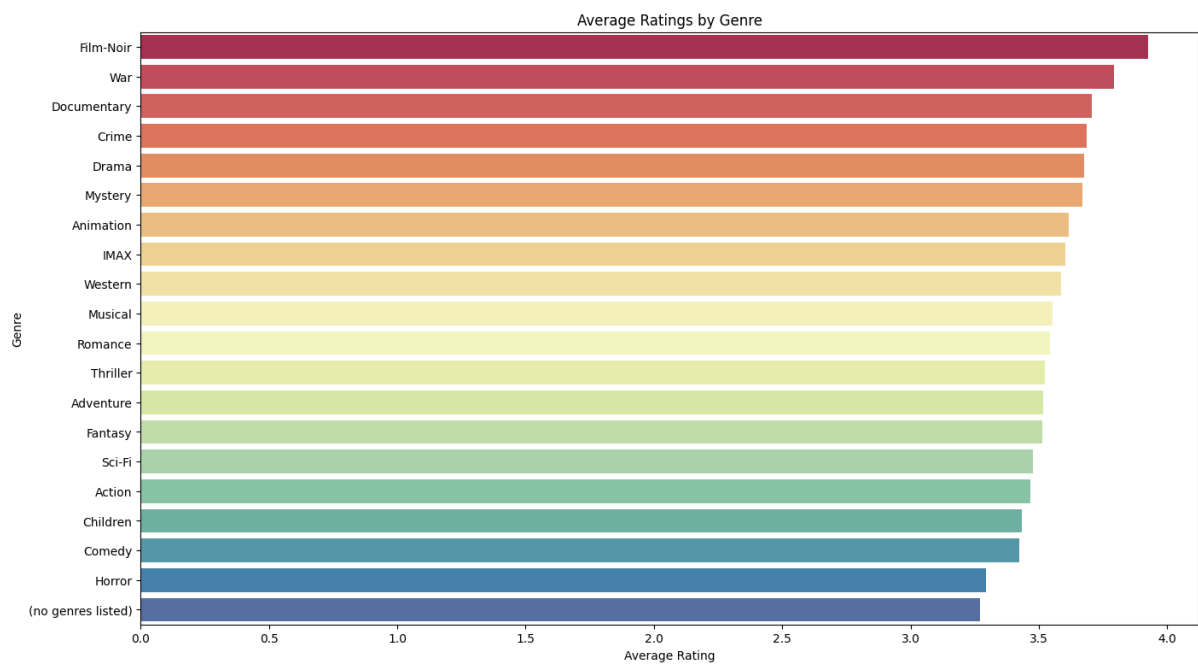


Figure 4.7 Average Ratings of Genres

4.5 Modelling

To make sure our system mitigates the echo chamber effect in the movie recommendation system, various models have used. The main goal of our system to provide users more accurate and diverse recommendations. To address the echo chamber issue, the system uses combination of content-based, collaborative, Singular Value Decomposition(SVD) and hybrid models.

4.5.1 Content Based Filtering Model

In this system the content-based filtering model uses the features of movie genres and release years. During pre-processing the genres of the movies are encoded using one-hot encoder and by feature engineering the release year of the movie is extracted from the movie title, then these features are merged together to form a feature set. Then the feature set is assigned with weights. The genre features are assigned with the weightage of “0.8” and the year feature is assigned with the weight of “0.2”. For providing more personalized recommendations, the genre of the movie is very important. So, by assigning the high weight to the genre feature, the model recommends movies that are very similar with the user’s genre preferences. The year feature of the movie is not as important as the genre feature, so by assigning a small weight to the year feature the model ensures that the year feature is not highly impacting the recommendation as the genre features. Even though the year feature is not highly influential like the genre feature, assigning a weight to the year feature introduces an element of diversity in the recommendations.

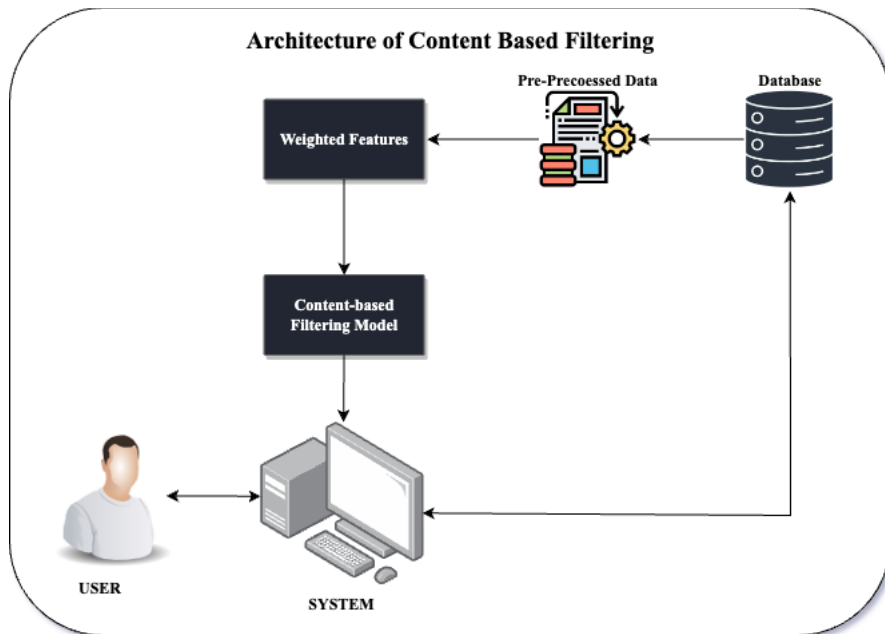


Figure 4.8 Architecture of Content Based Filtering Model

The “K-Nearest Neighbours” with “Cosine Similarity” as a metric is used in this model to provide recommendations. The “n_neighbors” parameter in the KNN is set to 100. Since we have “62,000”

movies in the dataset, if we keep the “n_neighbors” value as “62,000”, it will increase the computational cost of the system. So, for reducing the computational costs the value is set low. By using the cosine metric, the model calculates the similarity between the movies. When the model gets the information of the movie that the user have watched already, the model finds the movies that has the similar features as the input movie and recommends those movies. As this model provides recommendations based on the item’s similarity, it can reinforce the echo chamber issue if not combined with other models.

4.5.2 Collaborative Filtering Model

In this system the collaborative filtering model uses the “user_movie” matrix and the “user_similarity_matrix” to provide recommendations. The “user_movie” matrix shows the user rating to the movies. The “user_similarity_matrix” shows the similarity between the users. The similarity score is calculated using the cosine similarity.

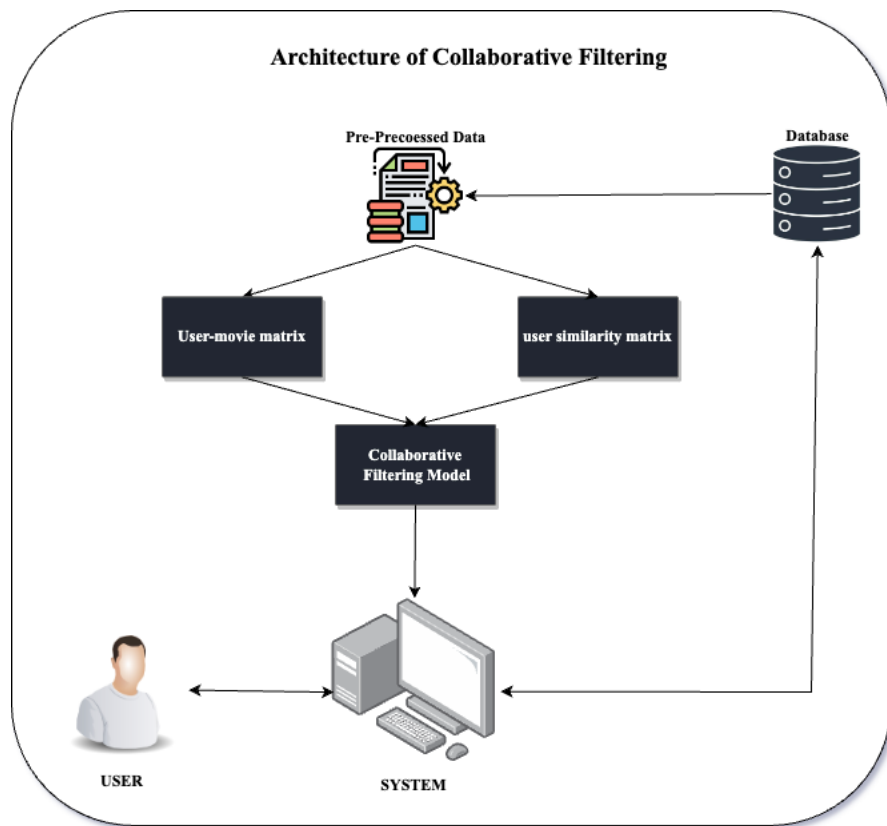


Figure 4.9 Architecture of Collaborative Filtering

For a given userId with the help of “user_movie” matrix and the “user_similarity_matrix”, the model selects the top N similar users to the given userId. Then the model recommends the movies that are top rated by the similar users but also not yet rated by the given user, to ensure that the recommendations does not contains the movies that the given user have already watched. As this model recommends movies based on the movies watched by the similar users, the model introduces diversity in the recommendations, but it does not considers the similarity between the movies. Also, this model

recommends the top-rated movies by the similar user, which might lead to the popularity bias. So, to mitigate the echo chamber issue this model should be combined with the other models to ensure the balance between the similarity and the diversity of the recommendations.

4.5.3 Singular Value Decomposition

In this system “Singular Value Decomposition” is used to predict the user’s rating for the movies that they have not rated already. The SVD model predicts the ratings that user would give to a movie, by using the matrix multiplication. By predicting the ratings of the movies, the model ranks these movies based on the predicted ratings. The SVD model creates a user-item matrix from the dataset. Then the whole dataset is used to create a training set, which ensures that the model accesses all the data during training. After training the model learn the latent factors for both the user and ratings to predict the user rating of unrated movies. Then for a given user, the model first finds out the unrated movies by the user, for each unrated movies the model predicts the rating and sorts them to identify the movies that the user would like. By using the SVD model the system understands more patterns about the data, resulting in providing more diverse recommendations but it still prefers the popular movies over the unpopular movies which will create the echo chamber issue. To mitigate the echo chamber issue, the SVD model should be combined with other models.

4.5.4 Hybrid Models

As all the models used before in the system will not mitigate the echo chamber issue when performed alone, the hybrid models are created. The Hybrid models in the system combine the strengths of different models used before in the system. The system has four different combination of hybrid models with different logics implemented in them, to increase its potential of mitigating the echo chamber issue.

4.5.4.1 Hybrid Model Variation-1

The “Hybrid Model-1” is designed by combining both the content-based model and the collaborative based model to provide more accurate set of recommendations. The main function of this hybrid model is to provide movie recommendations for a given userId and movie title. During the initialization of “Hybrid Model-1”, both the content-based filtering and collaborative filtering models are passed as arguments.

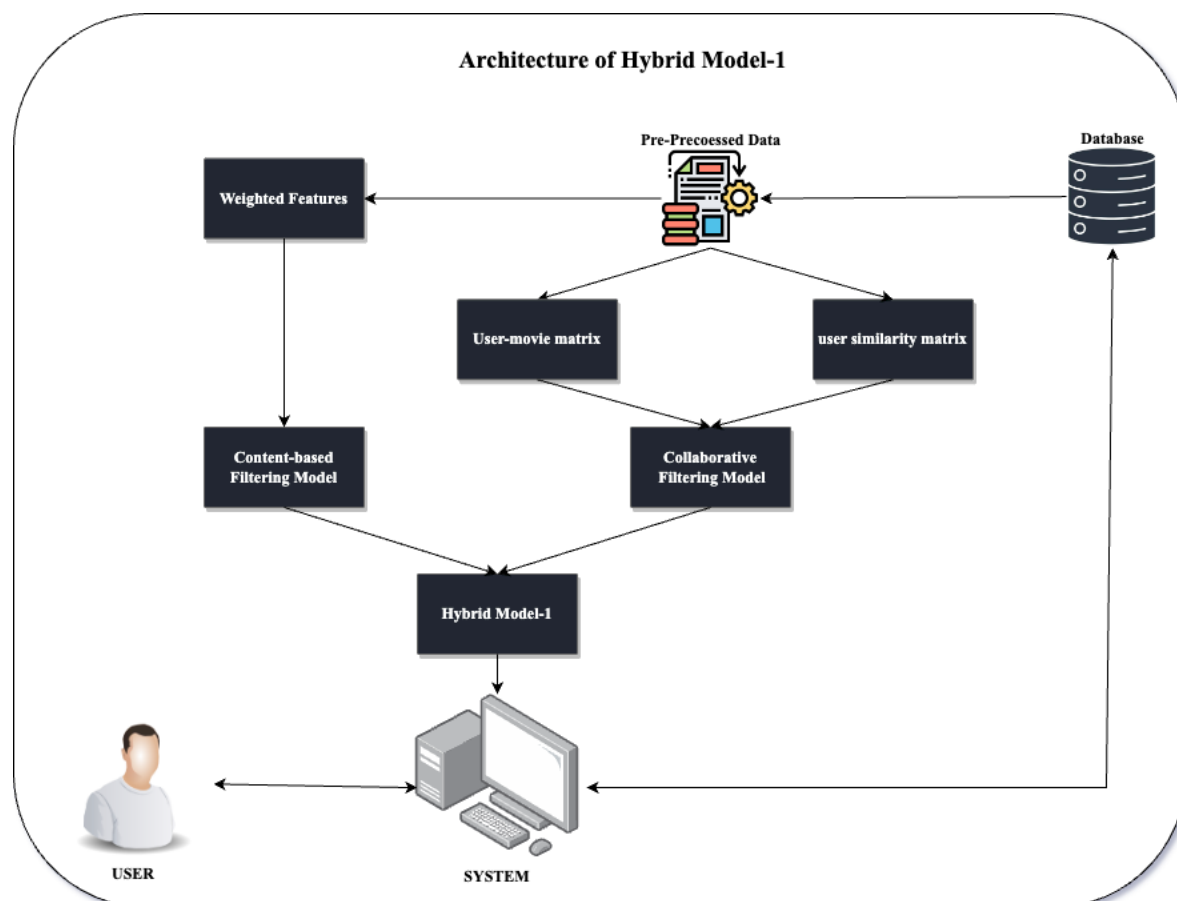


Figure 4.10 Architecture of Hybrid Model-1

For a given `userId` and movie title, the hybrid model gets the recommendations of both the content-based filtering model and collaborative filtering model. The recommendations from both these are combined into a single list. Then the list is sorted based on the “distance” value of the movies, which indicates the similarity of the recommendation. By combining both the content-based and collaborative based filtering models, the “Hybrid Model-1” introduces a level of diversity in the recommendation. The diversity is ensured as the content-based filtering recommends similar movies, and the collaborative filtering recommends movies liked by similar users.

4.5.4.2 Hybrid Model Variation-2

The “Hybrid Model-2” is a more advanced combination of both the content-based filtering and collaborative filtering models. Unlike the “Hybrid Model-1” this model not just combines the recommendations of both the filtering models, instead it has few more added logics to ensure serendipity and diversity in the recommendations. After getting recommendations from the filtering models the added logics comes into play to ensure that the recommendations are not only similar but also diverse.

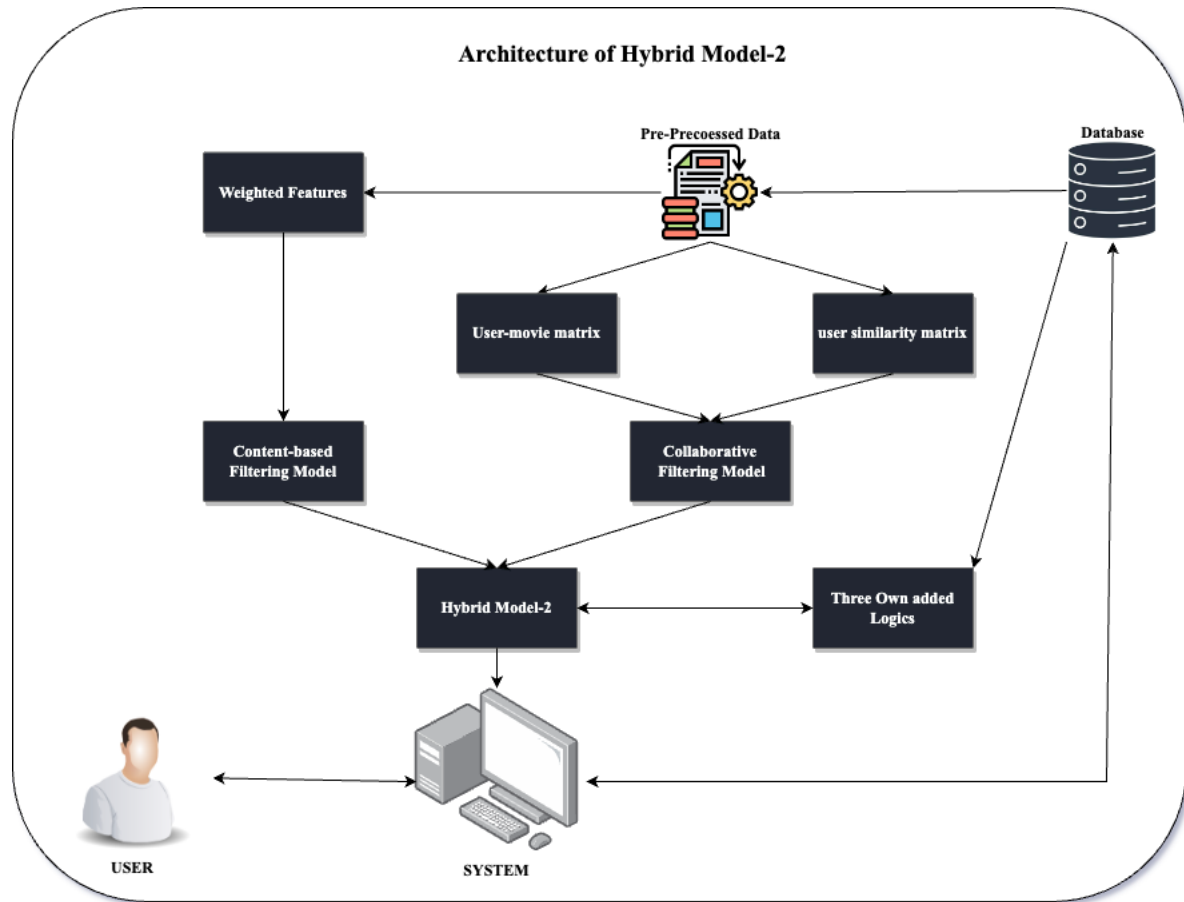


Figure 4.11 Architecture of Hybrid Model-2

After the “Hybrid Model-2” being initialized, for the given `userId` and the movie title it gets the recommendation from both the filtering models and combines them into a single list. After merging the recommendations, the “`add_serendipity`” function in this model replaces a certain percentage of the recommended movie with the non-recommended movies from the dataset. For the “`add_serendipity`” function the “`serendipity_percentage`” value is set to “0.2”, which is taken into consideration to decide how many movies are needed to be replaced in the recommendations. After the replacement of the recommended movies, the “`ensure_diversity`” function checks the distribution of genres in the recommendations. The “Threshold” value is set to “0.7”, if a genre’s distribution exceeds the threshold value, the “`ensure_diversity`” function reduces the number of movies recommended from that genre. This ensures that the recommendations are more diverse. After the “`add_serendipity`” and “`ensure_diversity`” functions, this model implements the “MMR” function, in which the function reorder the recommendations. The reordering is done by splitting the recommendation list into two sets, then the movies are selected alternatively from each set and the final list is sorted based on the distance score of the movie. This re-ordering ensures a balance between recommending movies that are more similar to the user’s preference. These added logics makes the model to address the echo chamber issue, by providing recommendations that are both similar to user’s taste and diverse in nature.

4.5.4.3 Hybrid Model Variation-3

The “Hybrid Model-3” is designed to combine strengths of both the content-based filtering model and the singular value decomposition model. The “Hybrid Model-3” provides user with a diverse set of movie recommendations. Unlike the above discussed models uses collaborative filtering, this model uses SVD model because due to computational challenges the collaborative filtering model uses a subset data which omitted the data of few movies, but the SVD model uses the whole dataset.

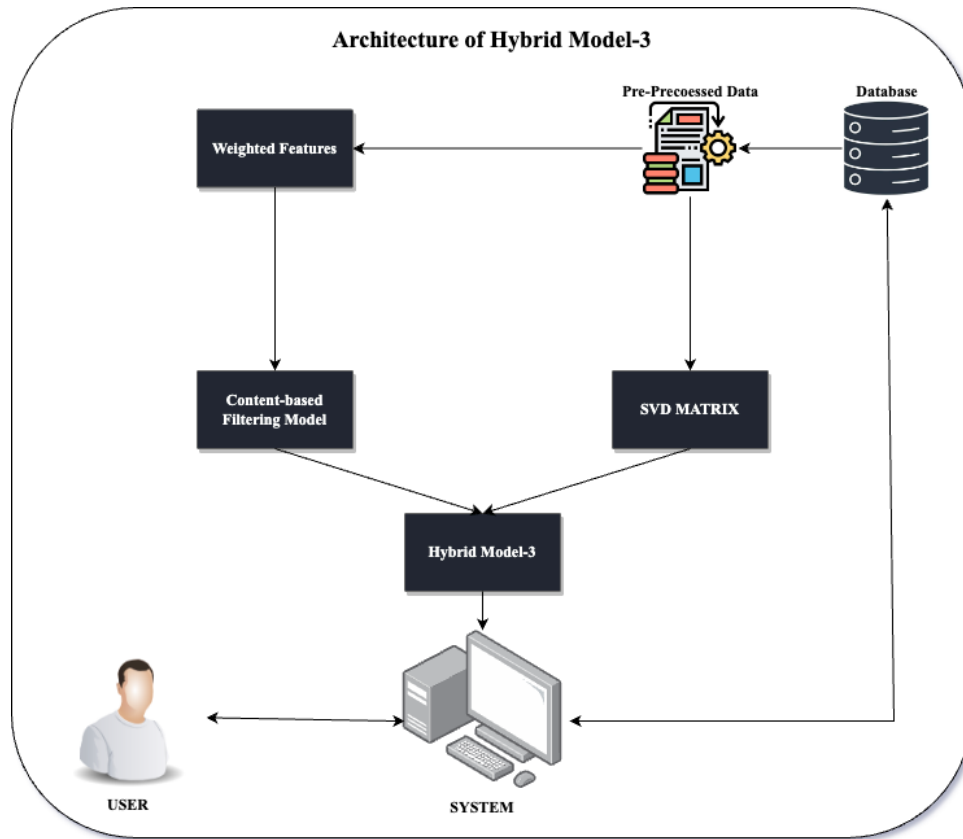


Figure 4.12 Architecture of Hybrid Model-3

After the “Hybrid Model-3” gets initialized, the “get_svd_recommendations” function starts fetching the recommendation made by the SVD model. The model also fetches the recommendations made by the content-based filtering model. Both these recommendations are combined into a single list. After combining them into a single list, the model drops the duplicates in the recommended list, as there might be few duplicated recommendations when combining the recommendations from two different models. Then the model provides the recommended movies to the user. By combining the two model’s recommendations, the model ensures that the user gets a mix of movies where few movies are similar to their preferences and few movies are recommended based on the popularity.

4.5.4.4 Hybrid Model Variation-4

The “Hybrid Model-4” is very advanced when comparing to the hybrid model-3, as it has few added logics to ensure the diversity and serendipity in the recommended set of movies. Same as the hybrid

model-3 this model combined the content-based model and the SVD model. The model tries to combine two different set of recommendation, where one of recommendation is the similar content movies to the provided input(content-based) and the other set of recommendation is the movies that are relevant to the input but also the recommendations are popular or relevant in the larger community(SVD).

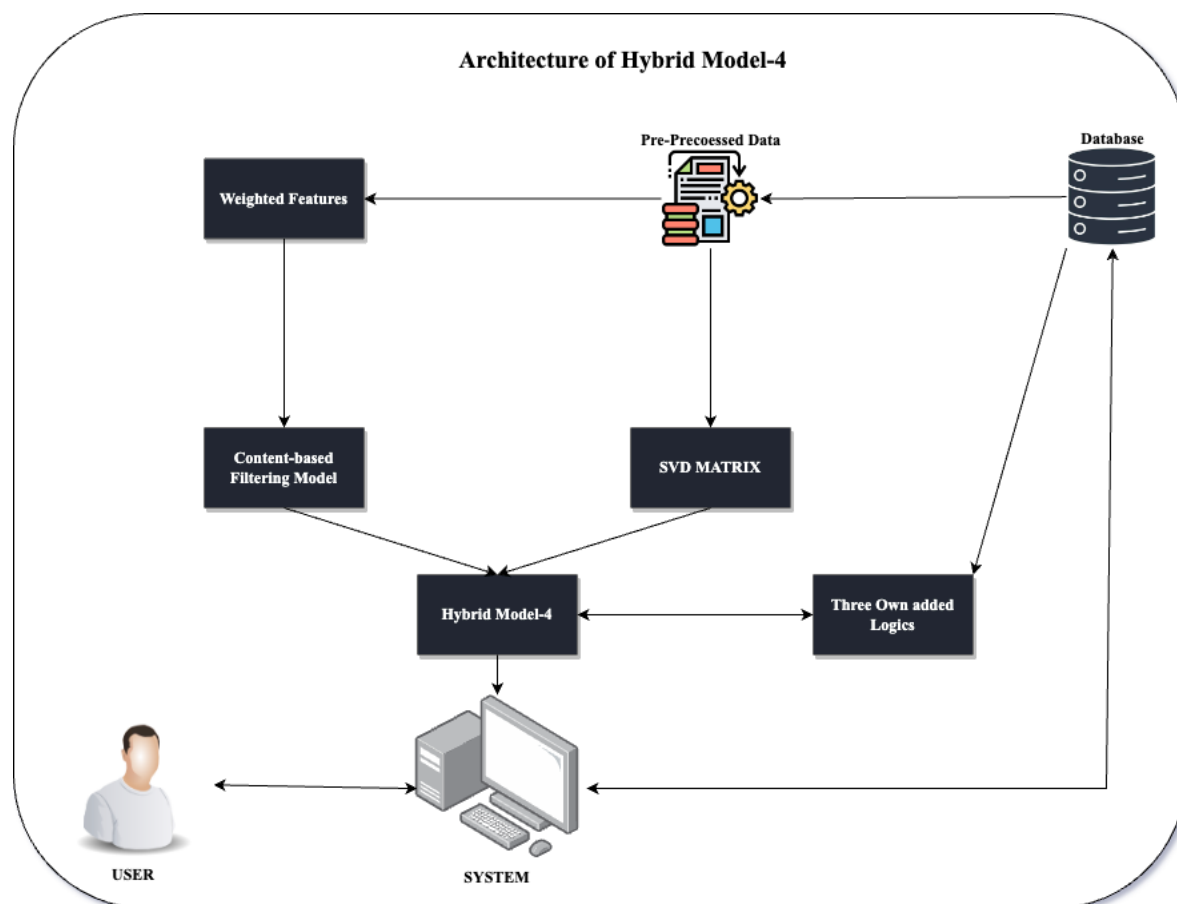


Figure 4.13 Architecture of Hybrid Model-4

When the model is initialized, with the given userId and movie title the model gets the recommendations from the SVD and content-based models and combines them into a single list. After combining the recommendations, the model removes the duplicated from the list. Then with the “serendipity_percentage” of “0.2” the “add_serendipity” function introduces an element of surprise by randomly replacing the recommended movies with the movies from the original dataset. Then with the “threshold” value of “0.7”, the “ensure_diversity” function checks if any distribution of genre in the recommended list is exceeding the threshold value. If it exceeds the threshold value, the function reduces the number of movies from that genre in the list. Then the “MMR” function alternatively selects the movies from the splitted list and sorts them based on the distance score of the movies. By combining the content-based model and the SVD model with the added logics, the “Hybrid Model-3” ensures that the users receives recommendations that are similar to their preferences and as well as with a percentage of diverse recommendations, which helps in making the user not get stuck in the echo chamber issue.

4.6 Evaluation Metrics

Since we've different variations of models in our system, evaluating the performance of the models is very important. The main reason to evaluate the models is to make sure that the models is providing relevant, diverse, and novel recommendations to the user. For the evaluation of the models in the system, four different types of evaluation metrics were used. Each of these evaluation metrics provides a unique performance of the models.

4.6.1 Genre Diversity

This metric calculates the diversity of the genres in the recommended list of movies. To mitigate the echo chamber issue, the model should ensure that it recommends diverse movies. This metric is calculated by analysing the distribution of genres in the recommended list and then the average is calculated. The high score of the genre diversity says that the recommended list of movies is more diverse, which can help in mitigating the echo chamber issue.

4.6.2 Similarity

This metric calculates the similarity between the recommended movies and the input movie. The Jaccard similarity co-efficient is used to calculate the similarity between genres of the recommended movies and genres of the input movie. For each recommended movie the Jaccard similarity co-efficient compares its genres with the genres of the input movie by the formula:

$$Jaccard(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

Where A and B are the genre sets of the input movie and the recommended movie[35]. After calculating the Jaccard similarity for all the movies in the recommendation list, the average of it is calculated. A high score of the Jaccard similarity indicates that the recommended movies shows that the recommended movies is closely related to the input movie.

4.6.3 Novelty

This Novelty metric calculates the score of how new or unexpected the recommended movies are. To mitigate the echo chamber issue, recommending unpopular movies is also important. This metric is calculated by the popularity of the recommended movies. In our system the popularity of the movie is calculated by the number of times the movie is rated. For each movie, its popularity is normalized by the maximum popularity value in the dataset. After calculating the novelty score for all the movies in the recommended lists, then the average of it is calculated.

4.6.4 Intra List similarity

The Intra list similarity metric calculates the similarity between movies between the movies in the recommended list. To calculate the Intra list similarity score the “Minkowski” distance is used. The “Minkowski” distance calculates the similarity of the feature vectors for all the movies in the recommended list. The distance of the feature vectors is converted into similarity using the formula:

$$\text{Similarity} = \frac{1}{1 + \text{Distance}}$$

After calculating the similarity score for all the movies in the recommendation list, the average of all the movies is calculated. The low score of Intra list similarity shows that the movies in the recommendation list are diverse from each other, which ensures that the model mitigates the echo chamber issue.

5 Results

5.1 Presentation of Results

Understanding the effectiveness and efficiency of the proposed models in the system is based on the quality and relevance of the recommendations made by the models. The results provided by the different models of the system are presented in this section, which gives the clear idea of what kind of recommendations each model gives.

movieid		title	genres	year
2203	2294	Antz (1998)	Adventure Animation Children Comedy Fantasy	1998
22285	114240	Aladdin (1992)	Adventure Animation Children Comedy Fantasy	1992
3021	3114	Toy Story 2 (1999)	Adventure Animation Children Comedy Fantasy	1999
3912	4016	Emperor's New Groove, The (2000)	Adventure Animation Children Comedy Fantasy	2000
3653	3754	Adventures of Rocky and Bullwinkle, The (2000)	Adventure Animation Children Comedy Fantasy	2000
9949	33463	DuckTales: The Movie - Treasure of the Lost La...	Adventure Animation Children Comedy Fantasy	1990
4780	4886	Monsters, Inc. (2001)	Adventure Animation Children Comedy Fantasy	2001
17430	91355	Asterix and the Vikings (Astérix et les Viking...	Adventure Animation Children Comedy Fantasy	2006
28227	131248	Brother Bear 2 (2006)	Adventure Animation Children Comedy Fantasy	2006
10773	45074	Wild, The (2006)	Adventure Animation Children Comedy Fantasy	2006
11604	53121	Shrek the Third (2007)	Adventure Animation Children Comedy Fantasy	2007
12969	65577	Tale of Despereaux, The (2008)	Adventure Animation Children Comedy Fantasy	2008
22633	115879	Toy Story Toons: Small Fry (2011)	Adventure Animation Children Comedy Fantasy	2011
22632	115875	Toy Story Toons: Hawaiian Vacation (2011)	Adventure Animation Children Comedy Fantasy	2011
23211	117454	The Magic Crystal (2011)	Adventure Animation Children Comedy Fantasy	2011
30443	136361	Scooby-Doo! Mask of the Blue Falcon (2012)	Adventure Animation Children Comedy Fantasy	2012
20014	103755	Turbo (2013)	Adventure Animation Children Comedy Fantasy	2013
57234	196693	Dragons: Dawn Of The Dragon Racers (2014)	Adventure Animation Children Comedy Fantasy	2014
22352	114552	Boxtrolls, The (2014)	Adventure Animation Children Comedy Fantasy	2014
30321	136016	The Good Dinosaur (2015)	Adventure Animation Children Comedy Fantasy	2015

Figure 5.1 Recommendations by Content-based Filtering Model

The figure 5.1 shows the recommendations made by the content-based model. The content-based model mainly focuses on the genres and the release year of the movie to make recommendations. So, for the

given movie title of “Toy Story (1995)” the content-based model recommends movies with the similar genres as the given input movie.

	title	movieId	genres	year
0	Avengers, The (2012)	89745	Action Adventure Sci-Fi IMAX	2012
1	Guardians of the Galaxy (2014)	112852	Action Adventure Sci-Fi	2014
2	The Hunger Games (2012)	91500	Action Adventure Drama Sci-Fi Thriller	2012
3	Pirates of the Caribbean: Dead Man's Chest (2006)	45722	Action Adventure Fantasy	2006
4	Thor (2011)	86332	Action Adventure Drama Fantasy IMAX	2011
5	Fifth Element, The (1997)	1527	Action Adventure Comedy Sci-Fi	1997
6	Iron Man 2 (2010)	77561	Action Adventure Sci-Fi Thriller IMAX	2010
7	V for Vendetta (2006)	44191	Action Sci-Fi Thriller IMAX	2006
8	I, Robot (2004)	8644	Action Adventure Sci-Fi Thriller	2004
9	Zombieland (2009)	71535	Action Comedy Horror	2009
10	Transformers (2007)	53996	Action Sci-Fi Thriller IMAX	2007
11	Deadpool (2016)	122904	Action Adventure Comedy Sci-Fi	2016
12	Now You See Me (2013)	102903	Crime Mystery Thriller	2013
13	Captain America: The First Avenger (2011)	88140	Action Adventure Sci-Fi Thriller War	2011
14	Captain America: The Winter Soldier (2014)	110102	Action Adventure Sci-Fi IMAX	2014
15	Mr. & Mrs. Smith (2005)	33679	Action Adventure Comedy Romance	2005
16	Kingsman: The Secret Service (2015)	119145	Action Adventure Comedy Crime	2015
17	Mummy, The (1999)	2617	Action Adventure Comedy Fantasy Horror Thriller	1999
18	Django Unchained (2012)	99114	Action Drama Western	2012
19	Star Trek (2009)	68358	Action Adventure Sci-Fi IMAX	2009

Figure 5.2 Recommendations by Collaborative Filtering Model

The figure 5.2 shows the recommendations made by the collaborative model. The collaborative filtering model recommends movies by finding out the movies watched by the similar users. For the given userId of “847” the collaborative filtering models recommends movies that are highly rated by the similar users of the given userId.

	title	movieId	genres	year
0	Antz (1998)	2294	Adventure Animation Children Comedy Fantasy	1998
1	Aladdin (1992)	588	Adventure Animation Children Comedy Musical	1992
2	Aladdin (1992)	114240	Adventure Animation Children Comedy Fantasy	1992
3	Toy Story 2 (1999)	3114	Adventure Animation Children Comedy Fantasy	1999
4	Emperor's New Groove, The (2000)	4016	Adventure Animation Children Comedy Fantasy	2000
5	Adventures of Rocky and Bullwinkle, The (2000)	3754	Adventure Animation Children Comedy Fantasy	2000
6	DuckTales: The Movie - Treasure of the Lost La...	33463	Adventure Animation Children Comedy Fantasy	1990
7	Monsters, Inc. (2001)	4886	Adventure Animation Children Comedy Fantasy	2001
8	Asterix and the Vikings (Astérix et les Viking...	91355	Adventure Animation Children Comedy Fantasy	2006
9	Brother Bear 2 (2006)	131248	Adventure Animation Children Comedy Fantasy	2006
10	Wild, The (2006)	45074	Adventure Animation Children Comedy Fantasy	2006
11	Shrek the Third (2007)	53121	Adventure Animation Children Comedy Fantasy	2007
12	Tale of Despereaux, The (2008)	65577	Adventure Animation Children Comedy Fantasy	2008
13	Toy Story Toons: Small Fry (2011)	115879	Adventure Animation Children Comedy Fantasy	2011
14	Toy Story Toons: Hawaiian Vacation (2011)	115875	Adventure Animation Children Comedy Fantasy	2011
15	The Magic Crystal (2011)	117454	Adventure Animation Children Comedy Fantasy	2011
16	Scooby-Doo! Mask of the Blue Falcon (2012)	136361	Adventure Animation Children Comedy Fantasy	2012
17	Turbo (2013)	103755	Adventure Animation Children Comedy Fantasy	2013
18	Dragons: Dawn Of The Dragon Racers (2014)	196693	Adventure Animation Children Comedy Fantasy	2014
19	Boxtrolls, The (2014)	114552	Adventure Animation Children Comedy Fantasy	2014

Figure 5.3 Recommendations by Hybrid Model-1

The figure 5.3 shows the first 20 recommendations made by the hybrid model 1. As the hybrid model-1 is the base hybrid model and it just combines the recommendations of both the filtering models, with the same userId “847” and same movie title “Toy Story (1995)” the recommended movies looks the same as both the content-based and collaborative filtering models.

	title	movielf	genres	year
0	Mask, The (1994)	367	Action Comedy Crime Fantasy	1994
1	Pagemaster, The (1994)	558	Action Adventure Animation Children Fantasy	1994
2	Fifth Element, The (1997)	1527	Action Adventure Comedy Sci-Fi	1997
3	Watership Down (1978)	2138	Adventure Animation Children Drama Fantasy	1978
4	Rush Hour (1998)	2273	Action Comedy Crime Thriller	1998
5	American Pie (1999)	2706	Comedy Romance	1999
6	Toy Story 2 (1999)	3114	Adventure Animation Children Comedy Fantasy	1999
7	We're Back! A Dinosaur's Story (1993)	3400	Adventure Animation Children Fantasy	1993
8	Emperor's New Groove, The (2000)	4016	Adventure Animation Children Comedy Fantasy	2000
9	Mummy Returns, The (2001)	4270	Action Adventure Comedy Thriller	2001
10	Shrek (2001)	4306	Adventure Animation Children Comedy Fantasy Ro...	2001
11	Atlantis: The Lost Empire (2001)	4366	Adventure Animation Children Fantasy	2001
12	Fast and the Furious, The (2001)	4369	Action Crime Thriller	2001
13	Land Before Time, The (1988)	4519	Adventure Animation Children Fantasy	1988
14	Flight of Dragons, The (1982)	5038	Adventure Animation Children Drama Fantasy	1982
15	Resident Evil (2002)	5219	Action Horror Sci-Fi Thriller	2002
16	Men in Black II (a.k.a. MIIB) (a.k.a. MIB 2) (...)	5459	Action Comedy Sci-Fi	2002
17	Pokemon 4 Ever (a.k.a. Pokémon 4: The Movie) (...)	5672	Adventure Animation Children Fantasy	2002
18	Sinbad: Legend of the Seven Seas (2003)	6536	Adventure Animation Children Fantasy	2003
19	Terminator 3: Rise of the Machines (2003)	6537	Action Adventure Sci-Fi	2003

Figure 5.4 Recommendations by Hybrid Model-2

The figure 5.4 shows the recommendations made by the hybrid model 2. The genres of the hybrid model-2 recommended movies are diverse and even the year of the movie is varied between each recommendation as the model focuses on ensuring diversity.

	title	movielf	genres	year
0	Antz (1998)	2294	Adventure Animation Children Comedy Fantasy	1998
1	Aladdin (1992)	588	Adventure Animation Children Comedy Musical	1992
2	Aladdin (1992)	114240	Adventure Animation Children Comedy Fantasy	1992
3	Toy Story 2 (1999)	3114	Adventure Animation Children Comedy Fantasy	1999
4	Emperor's New Groove, The (2000)	4016	Adventure Animation Children Comedy Fantasy	2000
5	Adventures of Rocky and Bullwinkle, The (2000)	3754	Adventure Animation Children Comedy Fantasy	2000
6	DuckTales: The Movie - Treasure of the Lost La...	33463	Adventure Animation Children Comedy Fantasy	1990
7	Monsters, Inc. (2001)	4886	Adventure Animation Children Comedy Fantasy	2001
8	Asterix and the Vikings (Astérix et les Viking...	91355	Adventure Animation Children Comedy Fantasy	2006
9	Brother Bear 2 (2006)	131248	Adventure Animation Children Comedy Fantasy	2006
10	Wild, The (2006)	45074	Adventure Animation Children Comedy Fantasy	2006
11	Shrek the Third (2007)	53121	Adventure Animation Children Comedy Fantasy	2007
12	Tale of Despereaux, The (2008)	65577	Adventure Animation Children Comedy Fantasy	2008
13	Toy Story Toons: Small Fry (2011)	115879	Adventure Animation Children Comedy Fantasy	2011
14	Toy Story Toons: Hawaiian Vacation (2011)	115875	Adventure Animation Children Comedy Fantasy	2011
15	The Magic Crystal (2011)	117454	Adventure Animation Children Comedy Fantasy	2011
16	Scooby-Doo! Mask of the Blue Falcon (2012)	136361	Adventure Animation Children Comedy Fantasy	2012
17	Turbo (2013)	103755	Adventure Animation Children Comedy Fantasy	2013
18	Dragons: Dawn Of The Dragon Racers (2014)	196693	Adventure Animation Children Comedy Fantasy	2014
19	Boxtrolls, The (2014)	114552	Adventure Animation Children Comedy Fantasy	2014

Figure 5.5 Recommendations by Hybrid Model-3

The figure 5.5 shows the recommendations made by the hybrid model 3. The hybrid model 3 just combines the recommendations made by the content-based filtering model and the singular value decomposition model, resulting in recommending the movies with the similar features of genre and year.

	title	movieid	genres	year
0	Three Colors: Red (Trois couleurs: Rouge) (1994)	306	Drama	1994
1	Shrek the Third (2007)	53121	Adventure Animation Children Comedy Fantasy	2007
2	Three Colors: Blue (Trois couleurs: Bleu) (1993)	307	Drama	1993
3	Niko & the Way to the Stars (a.k.a. The Flight...	62729	Adventure Animation Children Drama Fantasy	2008
4	Pagemaster, The (1994)	558	Action Adventure Animation Children Fantasy	1994
5	Lonely Wife, The (Charulata) (1964)	64241	Drama Romance	1964
6	Aladdin (1992)	588	Adventure Animation Children Comedy Musical	1992
7	Aladdin (1992)	114240	Adventure Animation Children Comedy Fantasy	1992
8	Tale of Despereaux, The (2008)	65577	Adventure Animation Children Comedy Fantasy	2008
9	Song of the Little Road (Pathar Panchali) (1955)	668	Drama	1955
10	Resan Till Melonia (1989)	69689	Adventure Animation Children Fantasy	1989
11	World of Apu, The (Apu Sansar) (1959)	670	Drama	1959
12	Partly Cloudy (2009)	72356	Animation Children Comedy Fantasy	2009
13	Vertigo (1958)	903	Drama Mystery Romance Thriller	1958
14	Duo (Pas de deux) (1968)	72915	Animation Musical	1968
15	Rear Window (1954)	904	Mystery Thriller	1954
16	Alone in the Wilderness (2004)	82143	Documentary	2004
17	Casablanca (1942)	912	Drama Romance	1942
18	Gnomeo & Juliet (2011)	84637	Adventure Animation Children Comedy Fantasy Ro...	2011
19	Sunset Blvd. (a.k.a. Sunset Boulevard) (1950)	922	Drama Film-Noir Romance	1950

Figure 5.6 Recommendations by Hybrid Model-4

The figure 5.6 shows the recommendations made by the hybrid model 4. The recommended movies in the hybrid model 4 have diverse genres and the movies are recommended from a wide range of release years as the models combines the recommendation for both the content-based model and singular value decomposition model and then implements the logics to ensure diversity.

Models	Genre Diversity	Similarity	Novelty	Intra List Similarity
Hybrid Model 2	0.8625	0.251344	0.142703	0.676653
Hybrid Model 3	0.4375	0.739095	0.028870	0.645762
Hybrid Model 4	0.8500	0.427699	0.109765	0.655149

Table 5.1 scores of Evaluation Metrics

The table 5.1 contains all the values of the four different evaluation metrics for all the variations of the hybrid models. The hybrid model 1 was not evaluated as it was the basic hybrid model which just combines the recommendations of content based and collaborative filtering models. From the table 5.1

it was observed that the Hybrid model 2 has the high score of genre diversity and novelty, the hybrid model 3 has the high score of similarity and low score of Intra list similarity and the hybrid model 4 has the low score of intra list similarity.

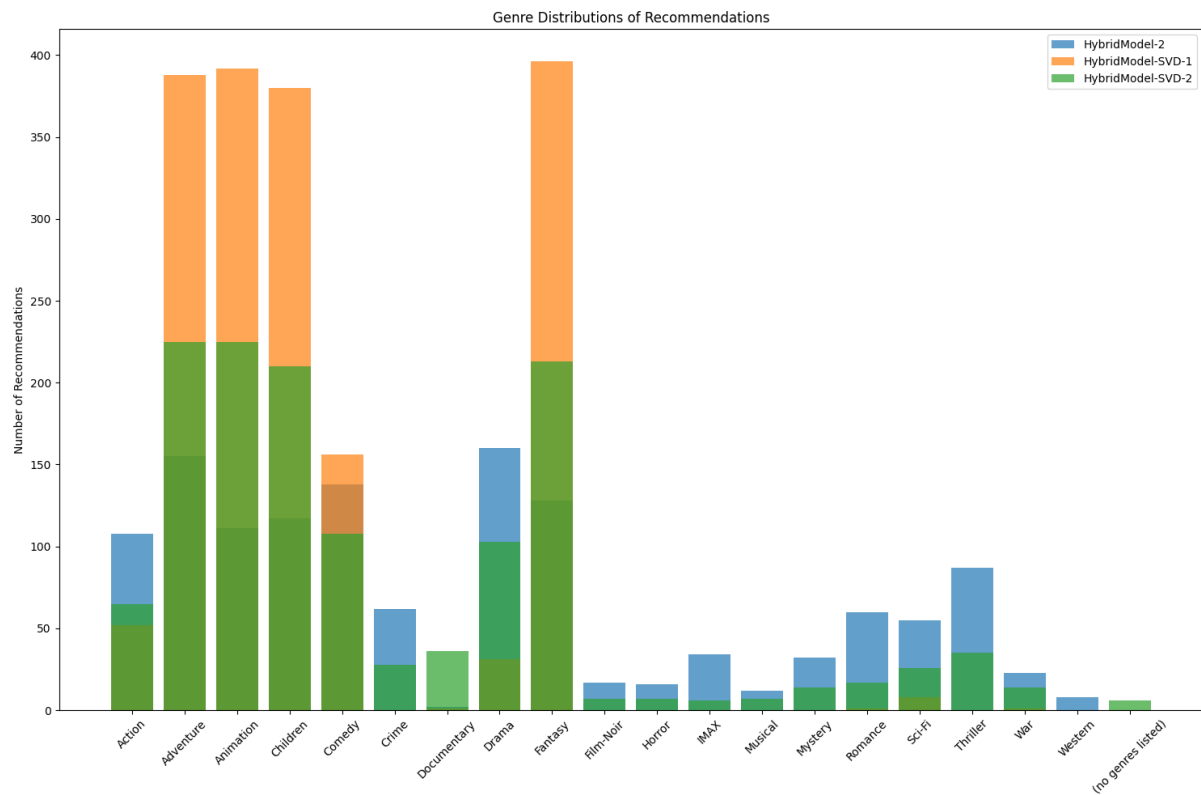


Figure 5.7 Genre distribution of Recommendations

The figure 5.7 shows the distribution of genres in the recommendation list of all three hybrid models. Understanding the distribution of genres in the recommendation list of the models, is important in evaluating the effectiveness and diversity of the recommendations.

6 Critical Review

6.1 Critical Review of the Results

The results of different variations of the hybrid models in the system gives a detailed understanding of how effective the model is addressing the echo chamber effect. This section critically evaluates the results and compares the performance of the models.

movieid	title		genres	year
0	1	Toy Story (1995)	Adventure Animation Children Comedy Fantasy	1995

Figure 6.1 Information of the Input Movie

The figure 6.1 shows the information of the input movie “Toy Story (1995)” given to the models to get recommendations. From the figure 5.1, which has the recommendations made by the content-based filtering model, we can understand that the content-based model recommends movies that are more similar to the given input movie. As the model recommends the model only depends on the limit set of features like genres and year. Even though the content-based model provides more personalised recommendations to the user, by mostly recommending similar genre movies this model potentially creates the echo chamber issue. On the other hand, from the recommendations of the collaborative filtering shown in the figure 5.2 we can see that the genres of the recommended movies are very well diverse but not similar to the given input movie. So, to ensure that the system recommends both similar and diverse set of recommendations the different variations of the hybrid models are developed in the system.

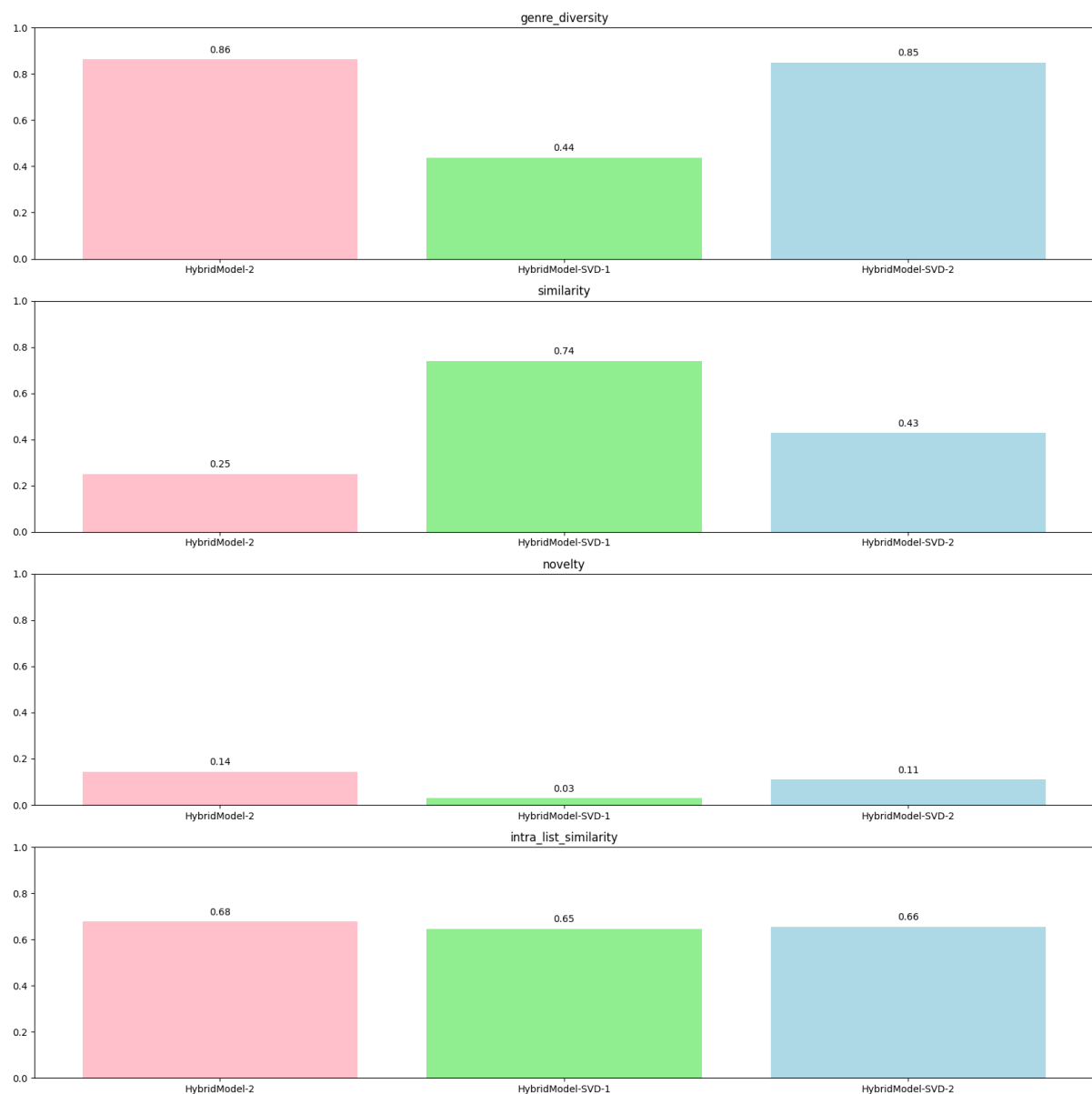


Figure 6.2 Visualization of the four Evaluation Metric Scores for the Three Main Hybrid Models

The figure 6.2 shows the four evaluation metric scores for the three main hybrid models excluding the hybrid model 1, as it was the basic hybrid model. The recommendations made by the hybrid model 1 as shown in the figure 5.3 shows that even after combining the recommendations of both the filtering models, the model still recommends most of the movies from the similar genres.

The Hybrid model 2 achieves the high score of the genre diversity metric, as it implements the “ensure_diversity”, “MMR” and “add_serendipity” functions after combining the recommendations of both the content-based and collaborative filtering models. But the hybrid model 2 achieves a moderate score of similarity metric as for the collaborative filtering the “user_movie” matrix and the “user_similarity_matrix” is taken from a subset of data. The novelty metric score of the hybrid model 2 is higher than the other two hybrid models, which shows that the model fails to recommend unpopular movies because for both the matrixes in collaborative filtering only the popular movies and active users are taken as a subset.

In hybrid model 3, the recommendations from the content-based model will have movies with the similar genres and the recommendations from the SVD model will have a diverse set of genres, because SVD recommends movies based on predicting the rating that the user might give. By combining both these results, the model achieves a below moderate score of the genre diversity metric. But the hybrid model 2 achieves a very high score of similarity metric, as it does not have any logics to introduce diversity and serendipity in the recommendations. The model 2 even achieves a very low score of novelty metric, which shows that the model is constantly recommending unpopular movies. The model recommends the unpopular movies as the SVD model is trained on the entire dataset.

In hybrid model 4, the recommendations are combined from the recommendations of both the content-based model and the SVD model. With the added logic of “add_serendipity”, “ensure_diversity” and “MMR” the model achieves a high score of genre diversity metric. Even with the added logics, the model 4 achieves a moderate score of the similarity metric. The model 4 achieves a better similarity metric score than the model 2 and lower than the model 3. As the model has logics to introduce diversity in the recommendation list, the model achieves a novelty score higher than the model 3 and lower than the model 2, which means the model recommends the unpopular movies better than the model 2 and lower than the model 3.

All the three models achieve a moderate score of Intra list similarity metric, which means all these models have nearly the same distribution of genres in the recommendations list. The intra list similarity score of these models ranges from “65%” to “68%”, which means all these models have an average of “66%” of similar genre movies in the recommendation list. This is due to the use of content-based filtering in all these models. By taking all the metrics into consideration, the Hybrid model 4 is found to be the most well balanced model to address the echo chamber issue in the system. The model ensures that, with the recommendation the user will get exposed to both relevant movies and new diverse movies. Hybrid model 2 provides great diversity, but as it achieves a high score of novelty showing that it recommends less unpopular movies, it limits the user only to the popular movies which might not

address the echo chamber issue in the system. The hybrid model 3 recommends most of the similar movies, which again does not address the echo chamber issue.

6.2 Critical Review of the Objective

Understanding the limitations of similar recommendations : The built system does a great job in achieving this objective. The use of content-based filtering, collaborative filtering and the Singular Value Decomposition shows the effort of the system to avoid the similar content recommendations. However, the real challenge in achieving this objective is to ensure the diverse recommendations, stays somewhat close to the user's preferences.

Ensuring Balanced User Exposure : The built hybrid models in the system, by combining both the content-based and collaborative filtering methods aims to find the balance between user preferences and diverse recommendations. The use of own logical functions like “add_serendipity”, “ensure_diversity” and “MMR” in the hybrid system shows the effort to introduce diversity in the recommendations. However, finding the correct balance is hard as the preferences varies from user to user.

Integration of own logic in the existing system : The system integrates the customs logics with the hybrid models to ensure the diversity in recommendations. The integrations of the logics with the hybrid models shows that the objective is achieved. However, the challenge in integrating the logic in the existing system, is that the logic integrated with the systems will not always predict the correct recommendations.

Providing continuous learning and adaptability : the built system does make use of the dynamic recommendation models, but the real-world application of continuously learning will require more data about the user behaviour to the recommendations, like click through rates and ratings to the recommended movies.

In conclusion, the objectives of the project and aim to mitigate the echo chamber issue is very challenging with the amount of data available. The system does makes efforts in achieving the objectives, but the real challenge is the practical application and ensuring the system continuously learning.

7 Conclusion

The 21st century have witnessed a huge need of the recommendation systems to provide user with the personalized recommendations. However, as technology develops the challenges also evolves with it.

One of the major problem in the recommendation system is the echo chamber issue, where the user get stuck inside a loop of getting only recommendations of similar contents. The main motive of this project is to mitigate the echo chamber issue in the movie recommendation system. To address the issue, this project tries to find the balance between providing the personalized recommendations along with diversity. To over the issue, the project proposes four different variations of hybrid models which combine the strengths of content based filtering, collaborative filtering, and singular value decomposition. Few own logics were also proposed in the project to ensure that the recommended movies are both personalized and diverse. In conclusion, by proposing four different variations of hybrid models, the project aims to provide more enhanced viewing experience to the users.

8 References

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